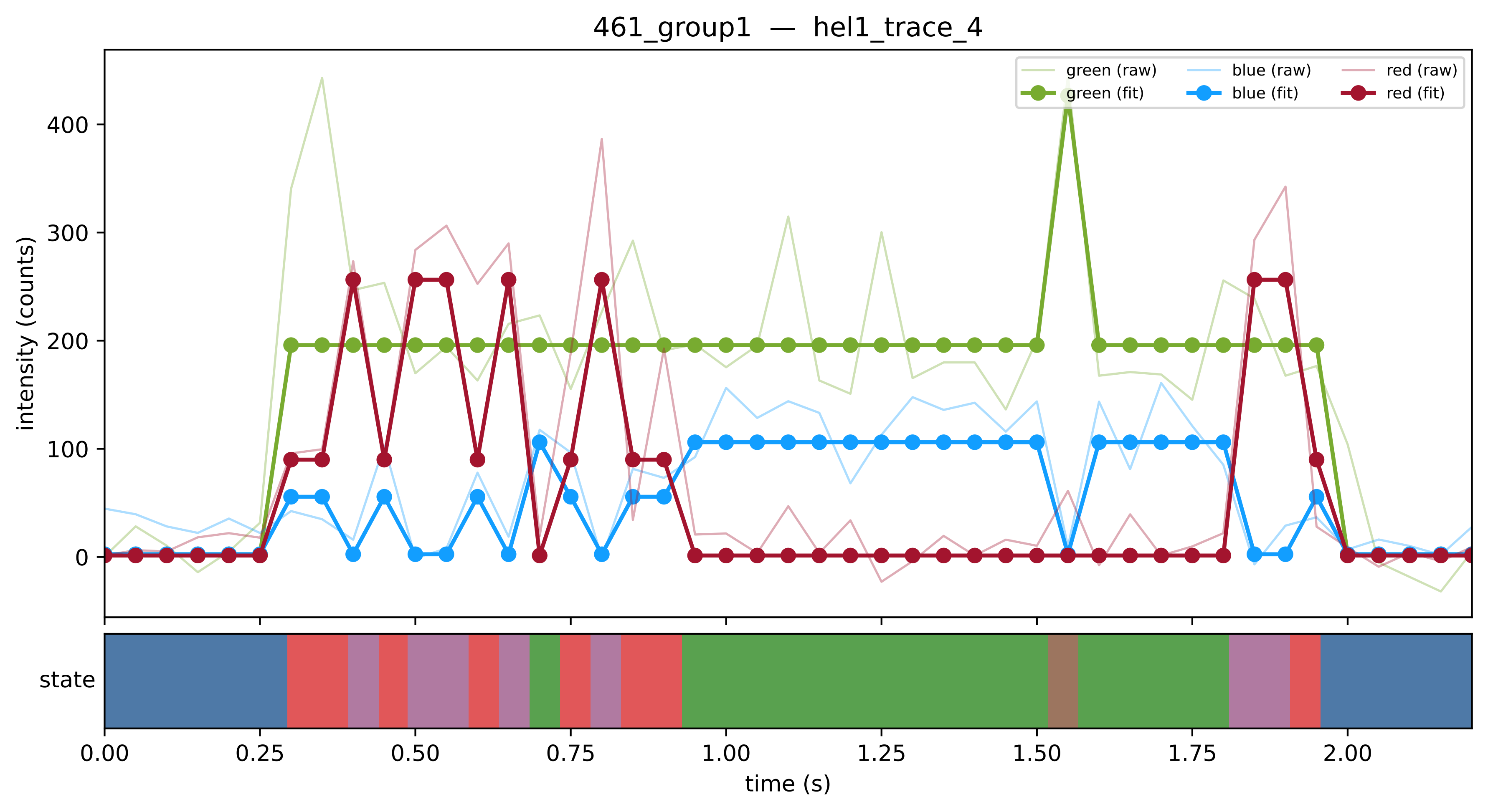
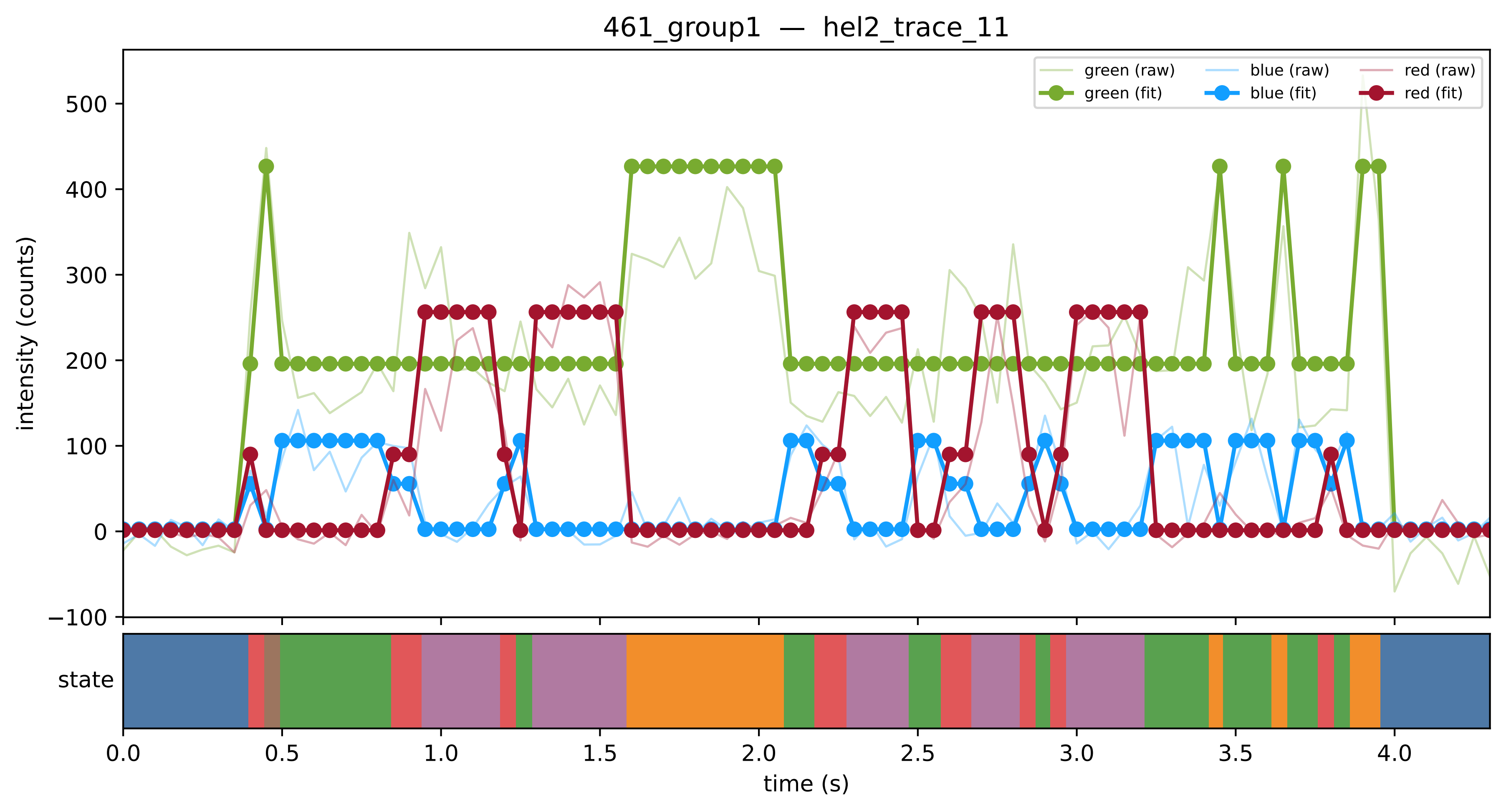
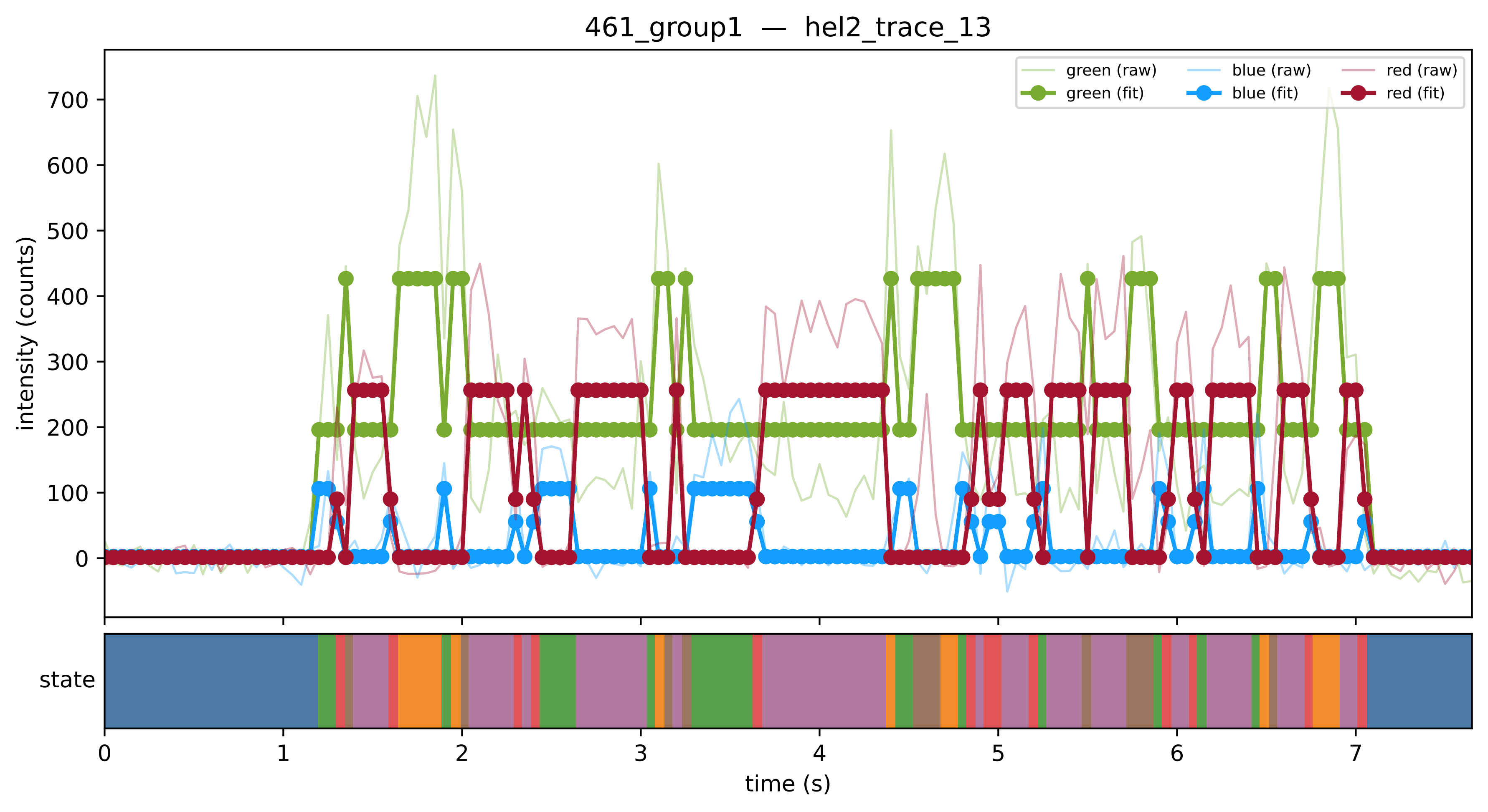
```
Condition: 461_group1
dt = 0.05 s

Transition probability matrix P (rows sum to 1):
   0.9353   0.0077   0.0109   0.0396   0.0052   0.0014
   0.0368   0.4402   0.2718   0.0930   0.0375   0.1208
   0.0118   0.0483   0.7195   0.1645   0.0436   0.0123
   0.0383   0.0236   0.3134   0.3552   0.2467   0.0227
   0.0048   0.0135   0.0526   0.2509   0.6293   0.0489
   0.0060   0.1402   0.1407   0.1658   0.3688   0.1785

Rate matrix K = logm(P)/dt   (units: 1/s)
K[i,j] for i!=j is the rate from state i to state j;
diagonal entries are negative total exit rates.
  -1.3762   0.2211  -0.1289   1.4638  -0.1657  -0.0142
   1.0426  -18.6510   8.5669   2.5467  -2.9552   9.4500
   0.0864   1.6746  -8.7339   7.1117  -0.2410   0.1022
   1.3656   0.3868   14.0214  -28.1806   11.7591   0.6477
  -0.1701  -0.1117  -1.7800   12.2189  -13.0270   2.8701
  -0.1853   10.5303   3.7214   4.1463   22.3661  -40.5787

Off-diagonal rate constants (state i -> state j, 1/s):
  k(0 -> 1) =  0.22113
  k(0 -> 2) = -0.12891
  k(0 -> 3) =  1.46381
  k(0 -> 4) = -0.16566
  k(0 -> 5) = -0.01417
  k(1 -> 0) =  1.04263
  k(1 -> 2) =  8.56691
  k(1 -> 3) =  2.54668
  k(1 -> 4) = -2.95523
  k(1 -> 5) =  9.45001
  k(2 -> 0) =  0.08637
  k(2 -> 1) =  1.67462
  k(2 -> 3) =  7.11170
  k(2 -> 4) = -0.24098
  k(2 -> 5) =  0.10218
  k(3 -> 0) =  1.36563
  k(3 -> 1) =  0.38683
  k(3 -> 2) =  14.02137
  k(3 -> 4) =  11.75912
  k(3 -> 5) =  0.64767
  k(4 -> 0) = -0.17011
  k(4 -> 1) = -0.11174
  k(4 -> 2) = -1.78004
  k(4 -> 3) =  12.21887
  k(4 -> 5) =  2.87006
  k(5 -> 0) = -0.18534
  k(5 -> 1) =  10.53034
  k(5 -> 2) =  3.72137
  k(5 -> 3) =  4.14627
  k(5 -> 4) =  22.36610
```



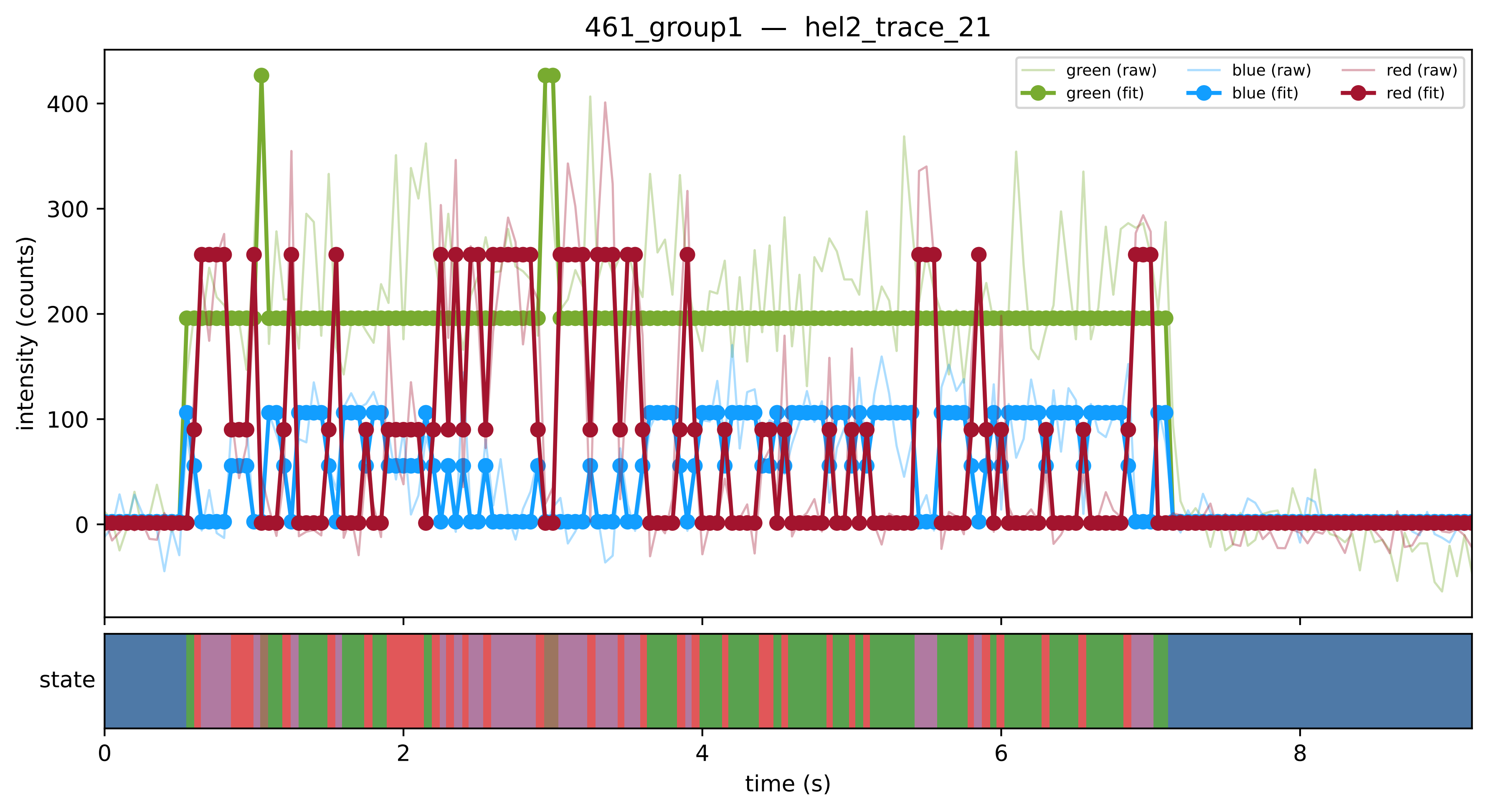
461_group1 — hel1_trace_4
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time (s)



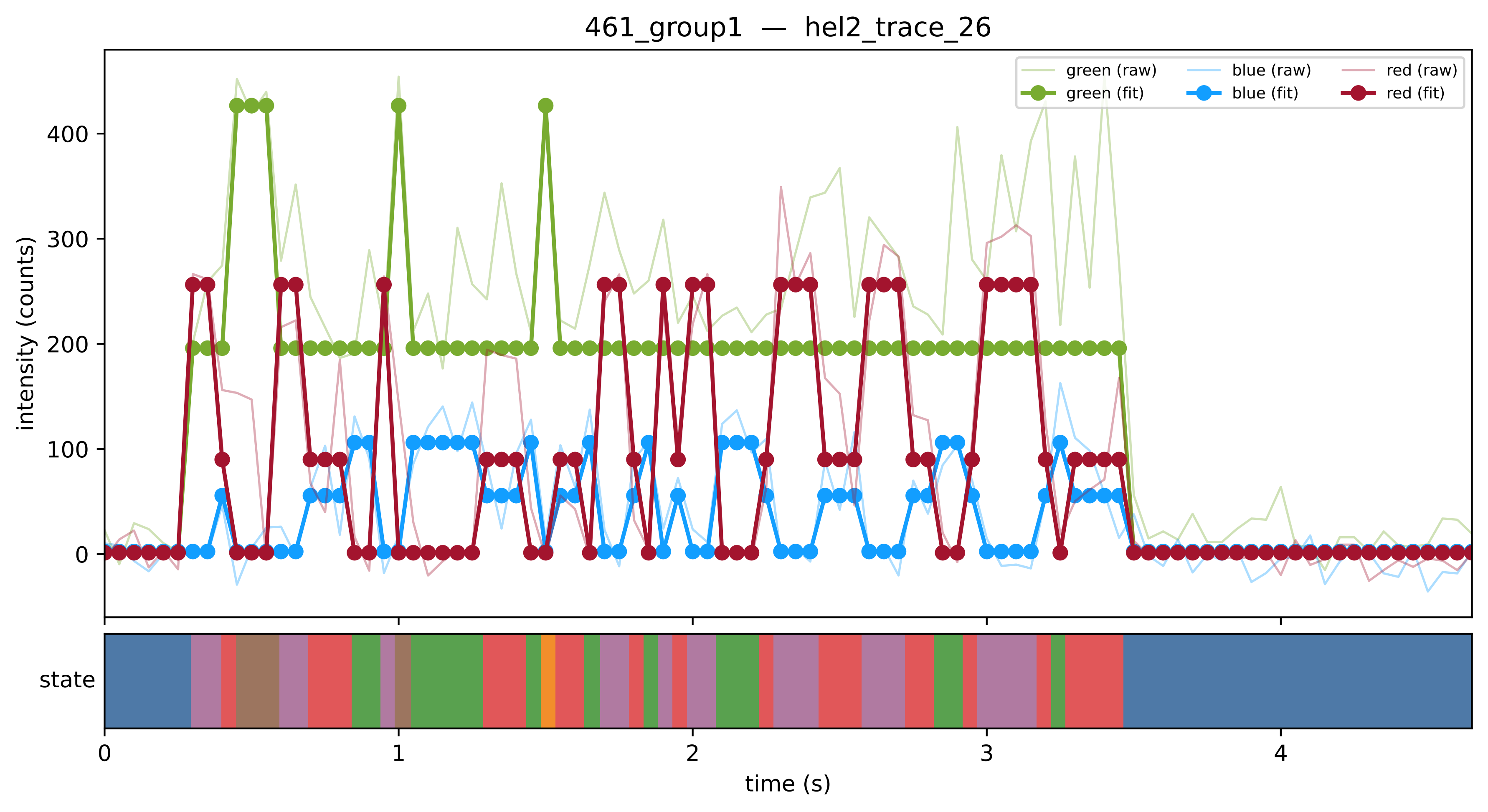
461_group1 — hel2_trace_11
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
500
400
300
200
100
0
−100
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
time (s)



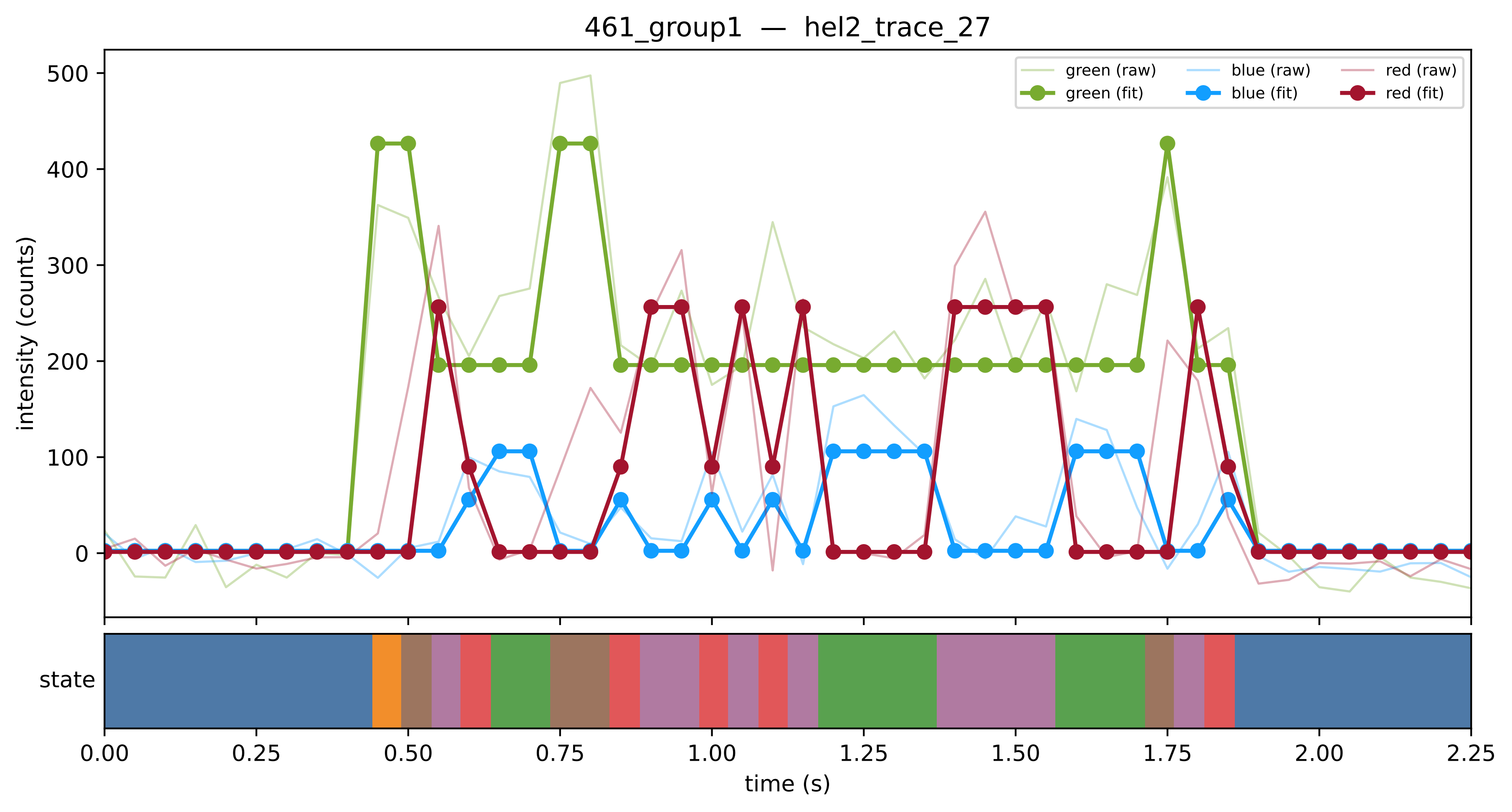
461_group1 — hel2_trace_13
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
500
600
700
state
0
1
2
3
4
5
6
7
time (s)

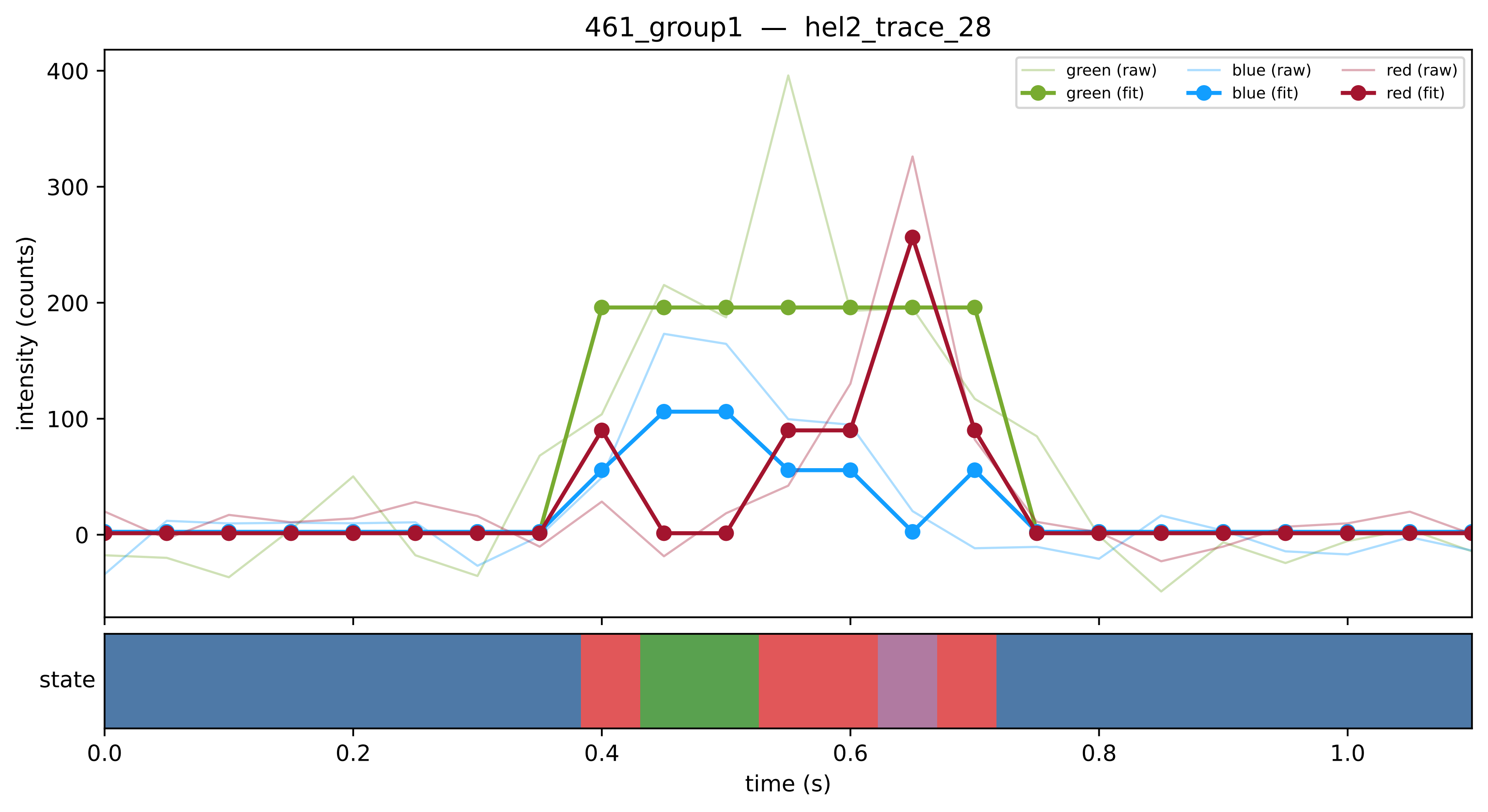


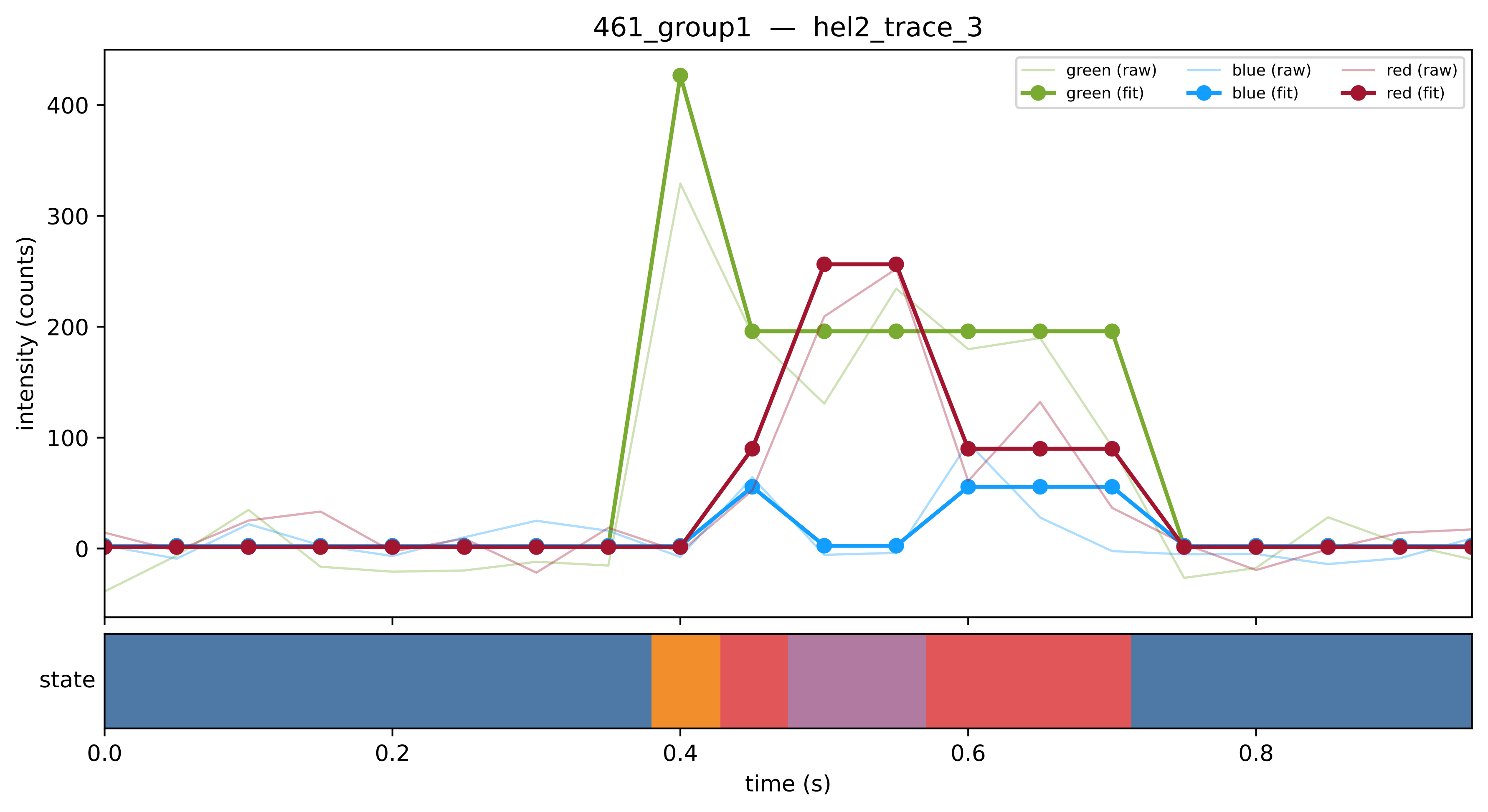
461_group1 — hel2_trace_21
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0
2
4
6
8
time (s)



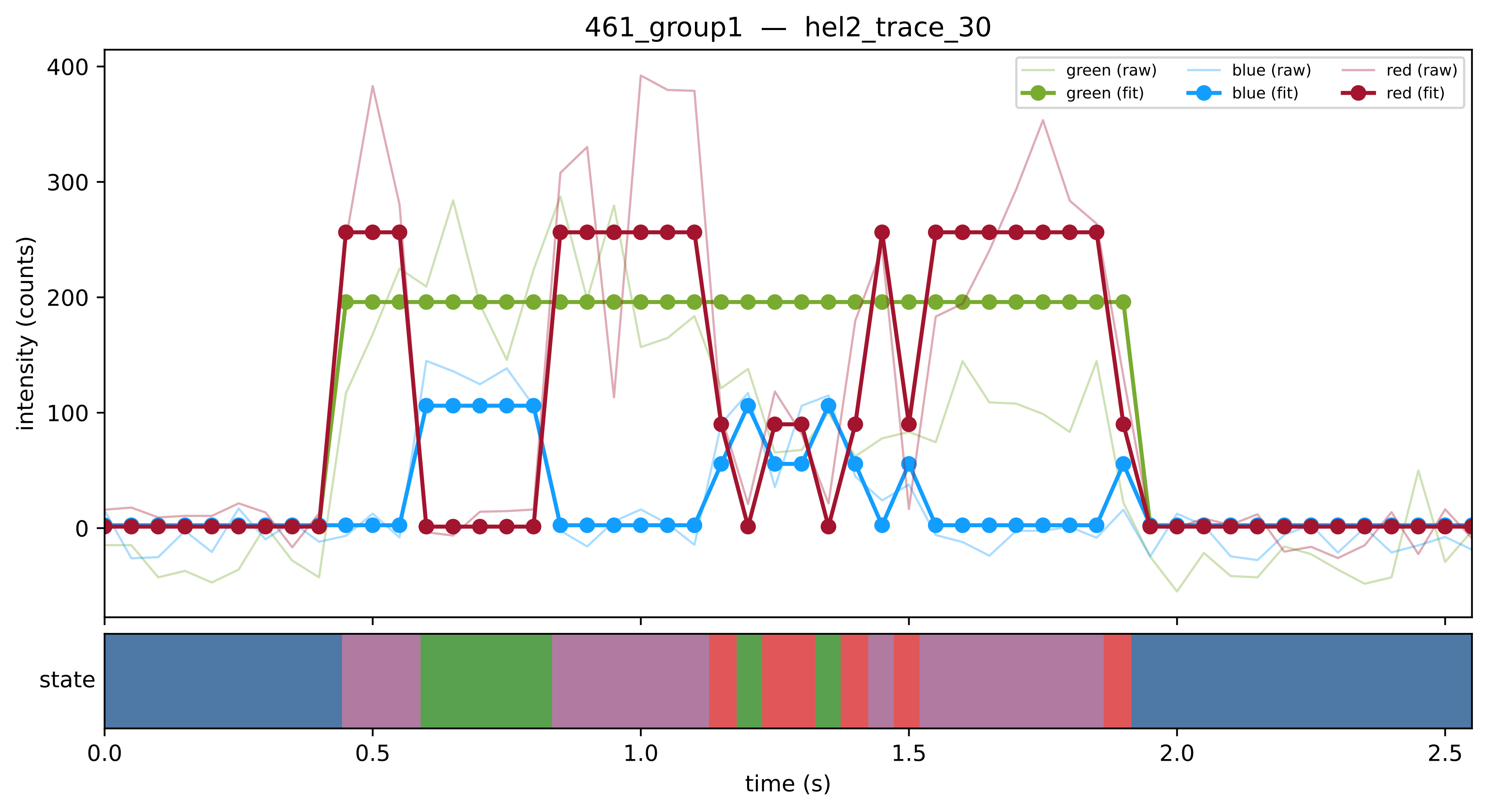
461_group1 — hel2_trace_26
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0
1
2
3
4
time (s)


461_group1 — hel2_trace_27
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
100
200
300
400
500
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25

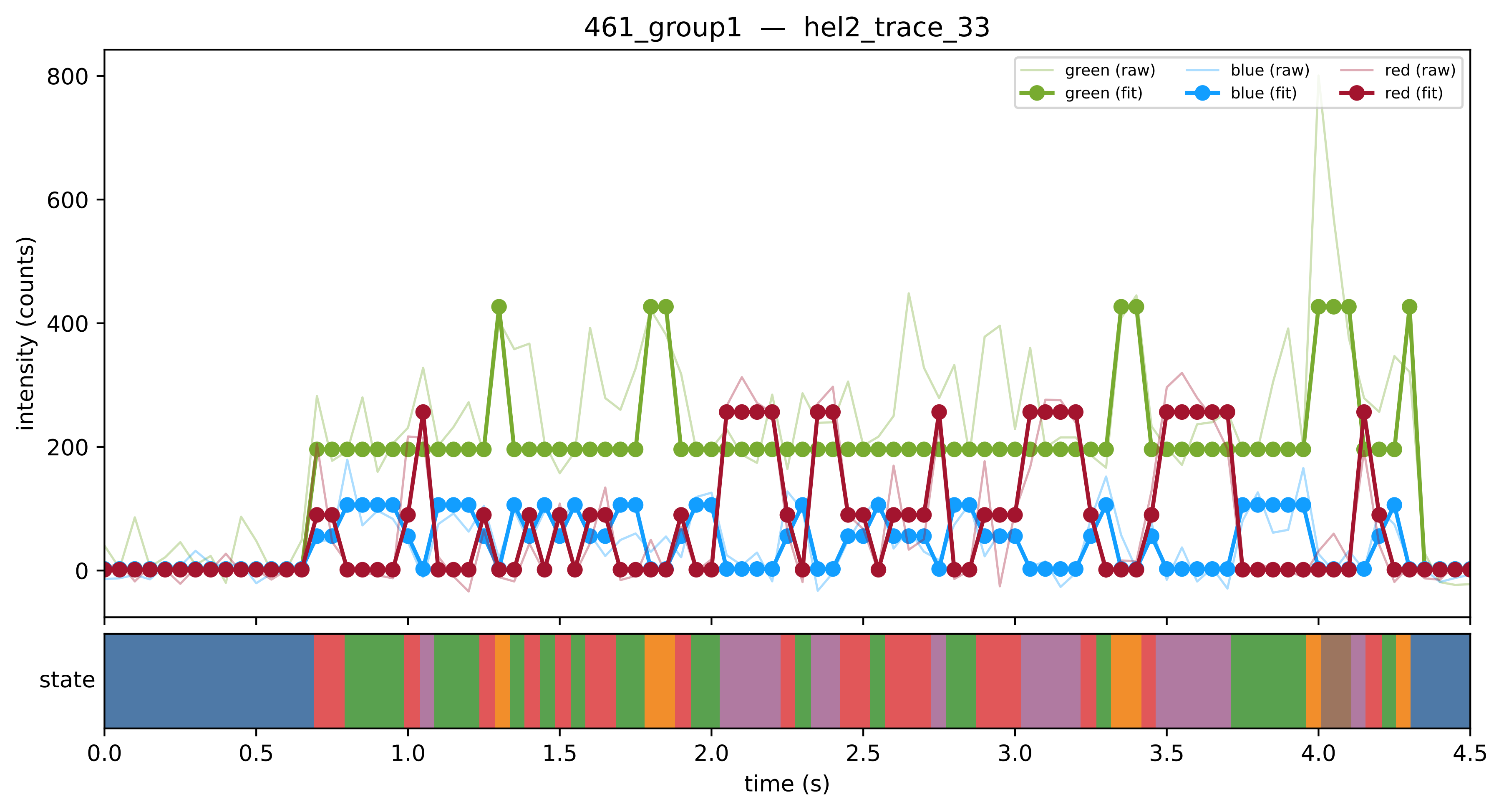

461_group1 — hel2_trace_28
intensity (counts)
400
300
200
100
0
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0.0
0.2
0.4
0.6
0.8
1.0
time (s)



461_group1 — hel2_trace_3
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0.0
0.2
0.4
0.6
0.8
time (s)

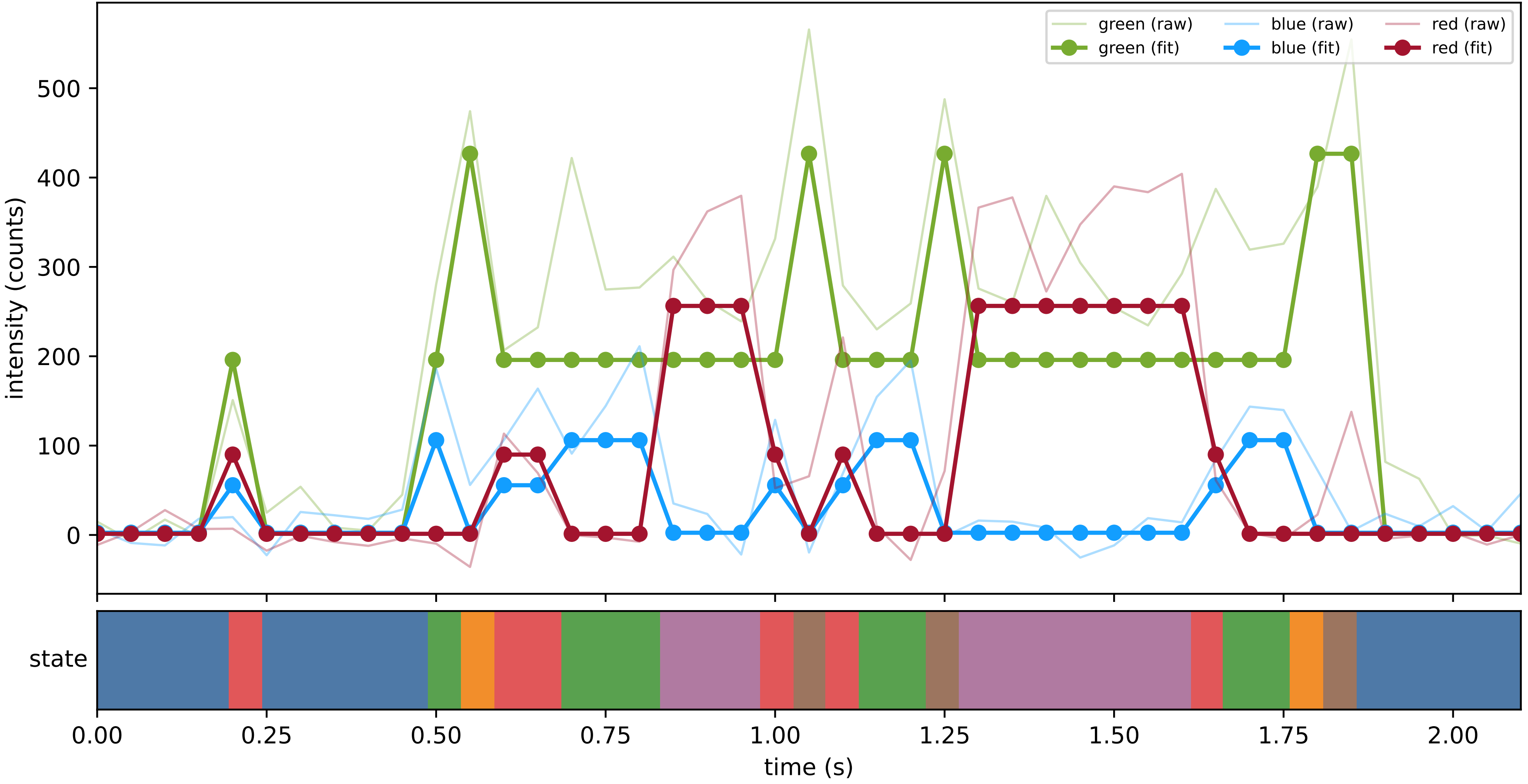


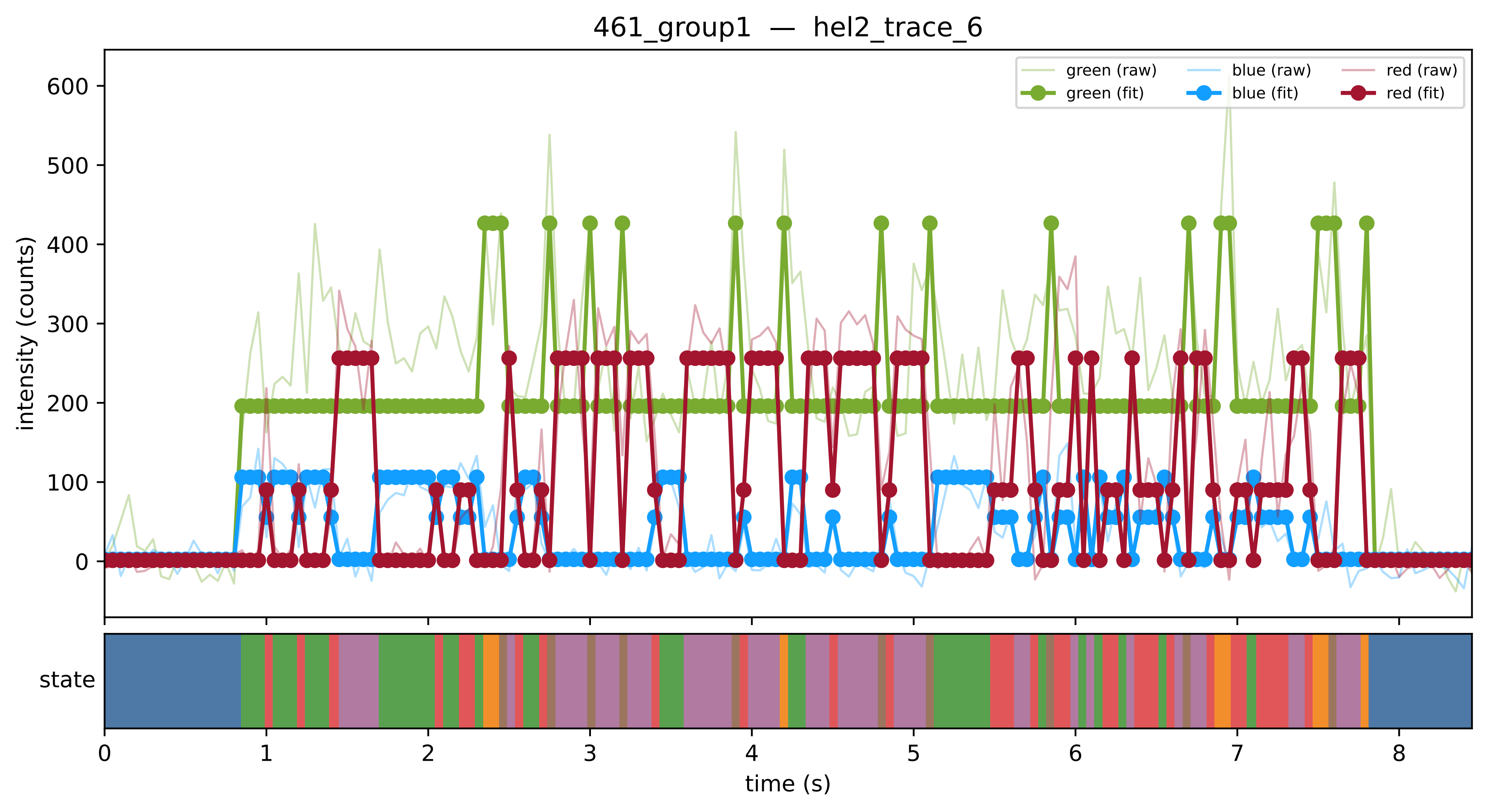
461_group1 — hel2_trace_30
intensity (counts)
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)



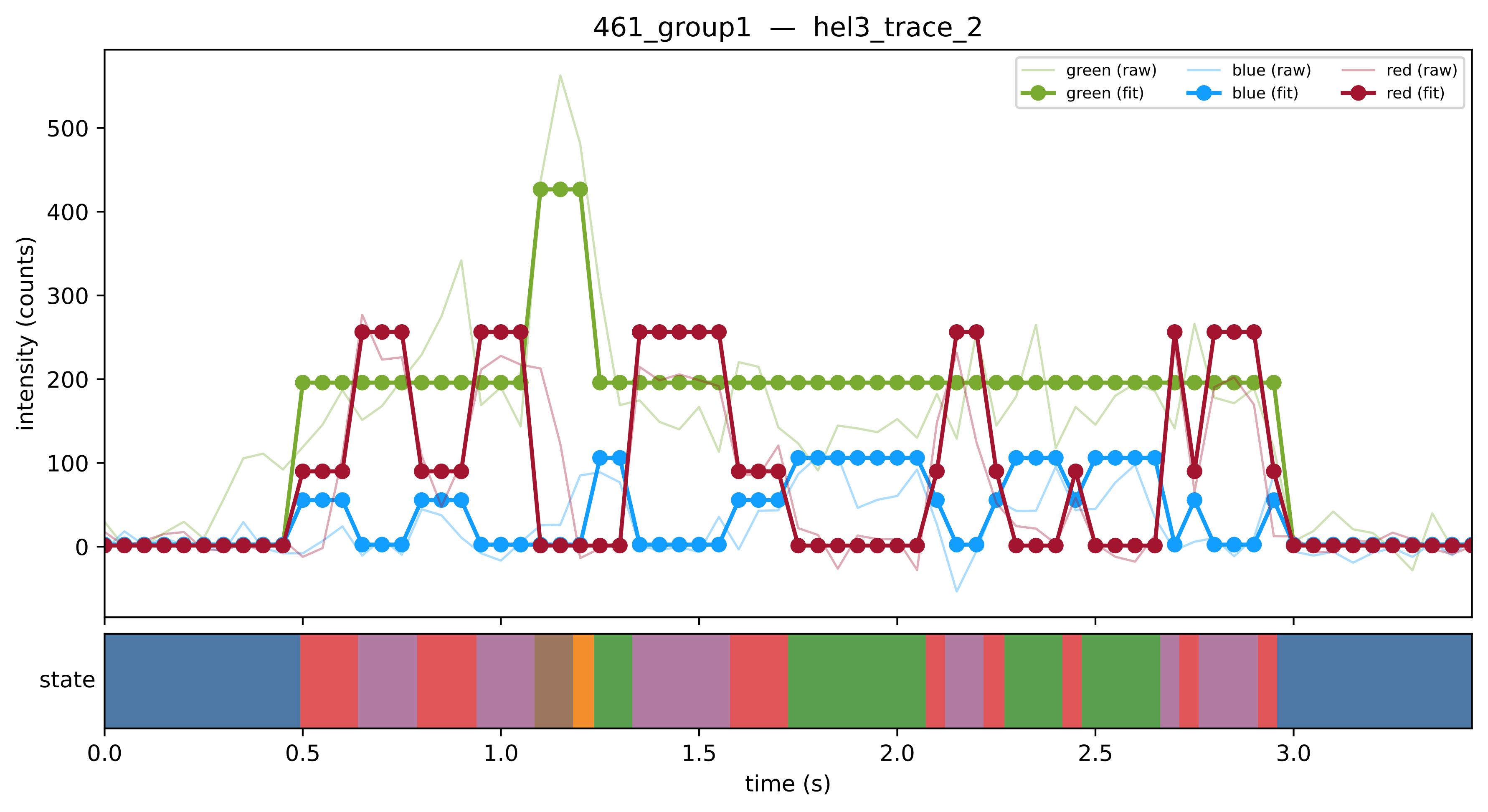
461_group1 — hel2_trace_33
intensity (counts)
800
600
400
200
0
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
time (s)

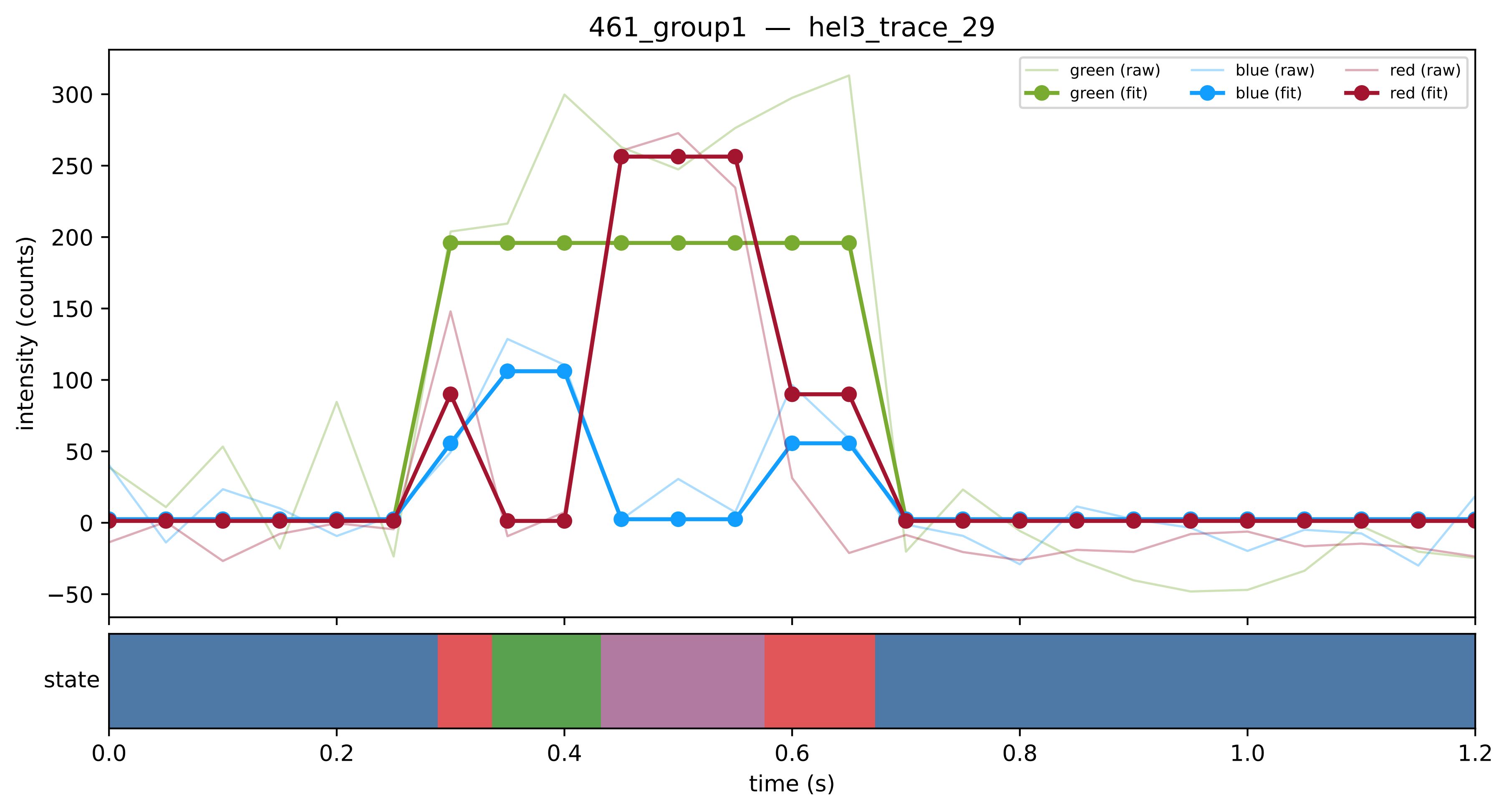
461_group1 — hel2_trace_46



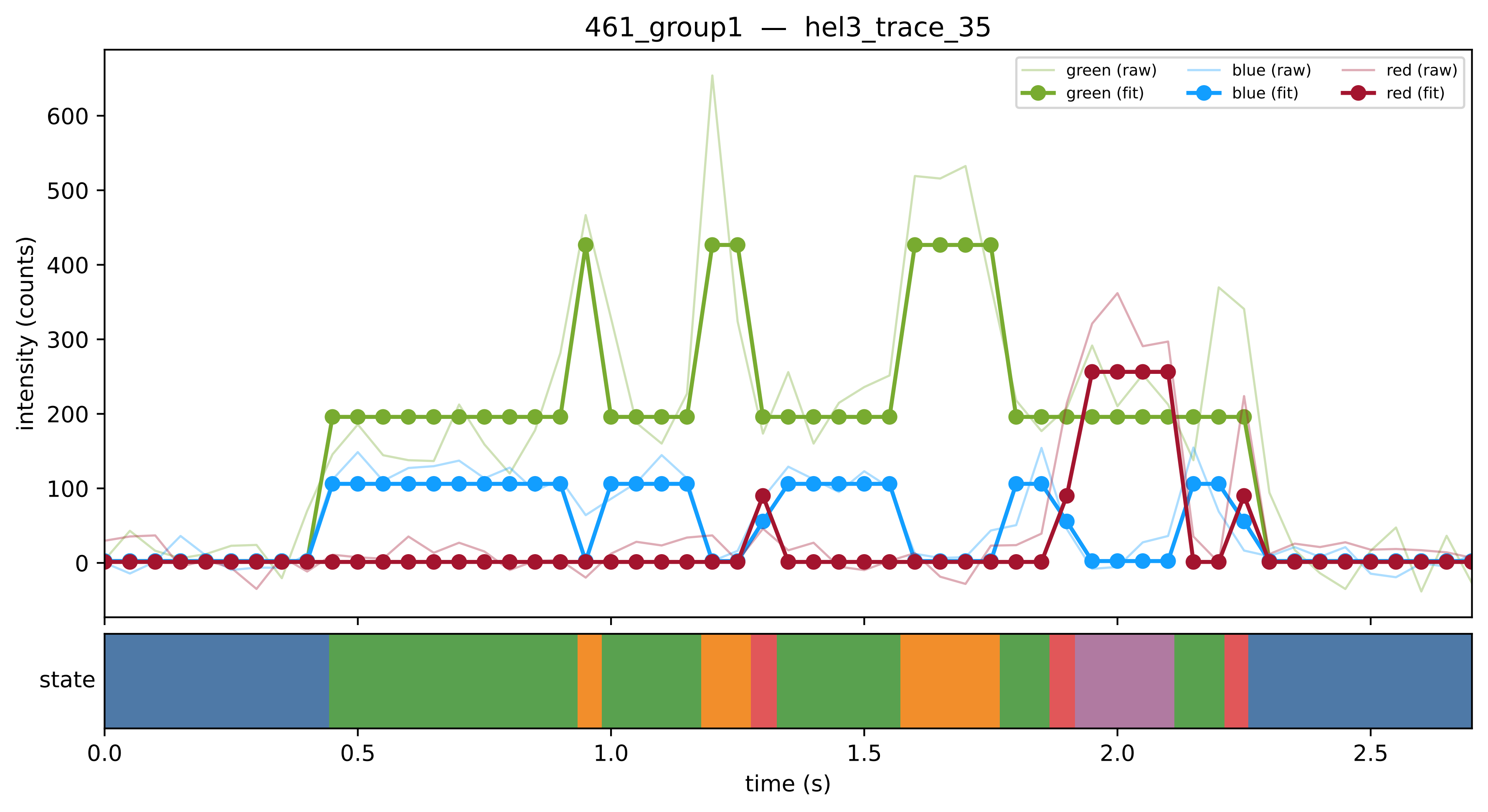


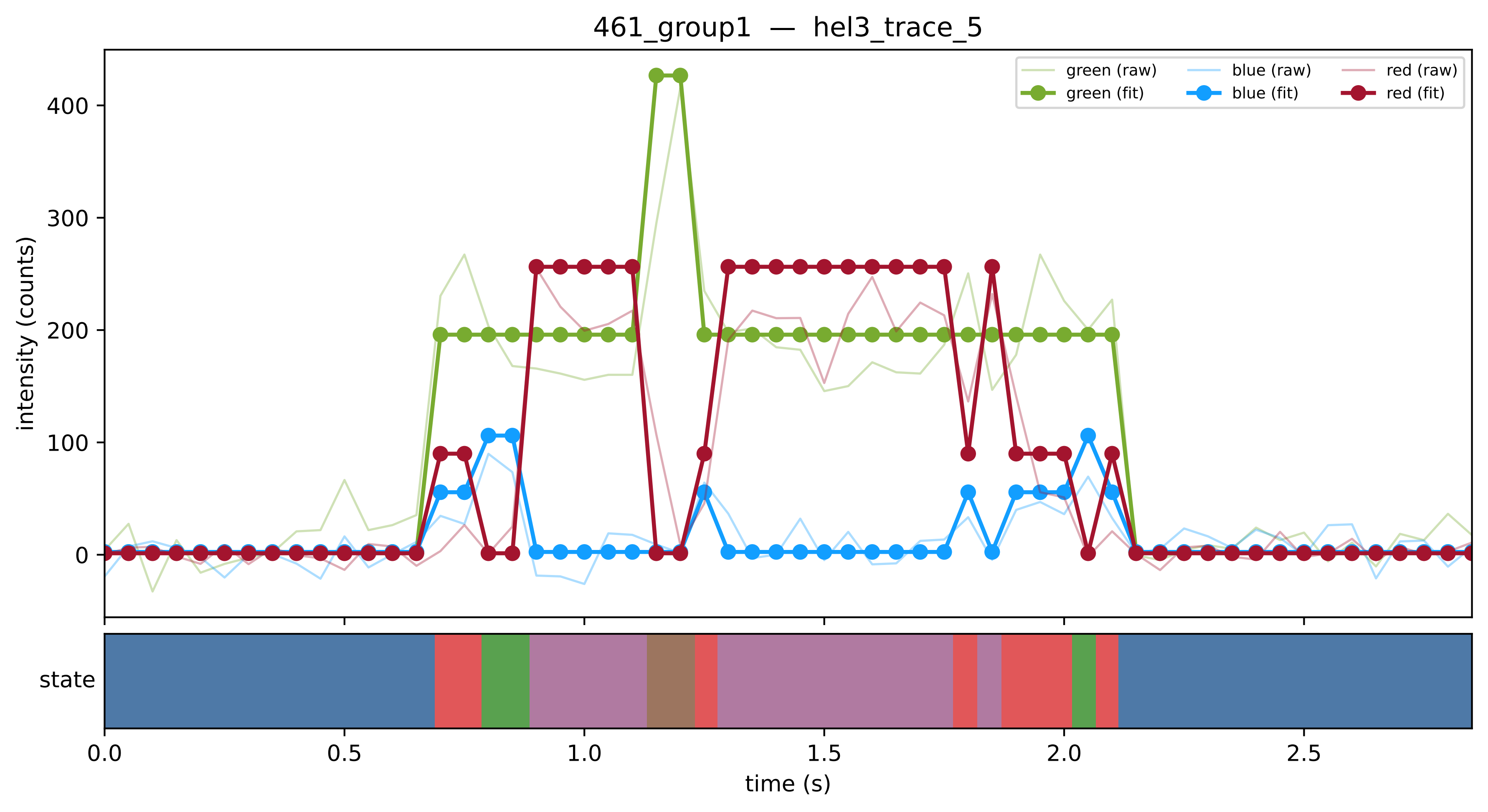
461_group1 — hel2_trace_6
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
600
500
400
300
200
100
0
state
0
1
2
3
4
5
6
7
8
time (s)

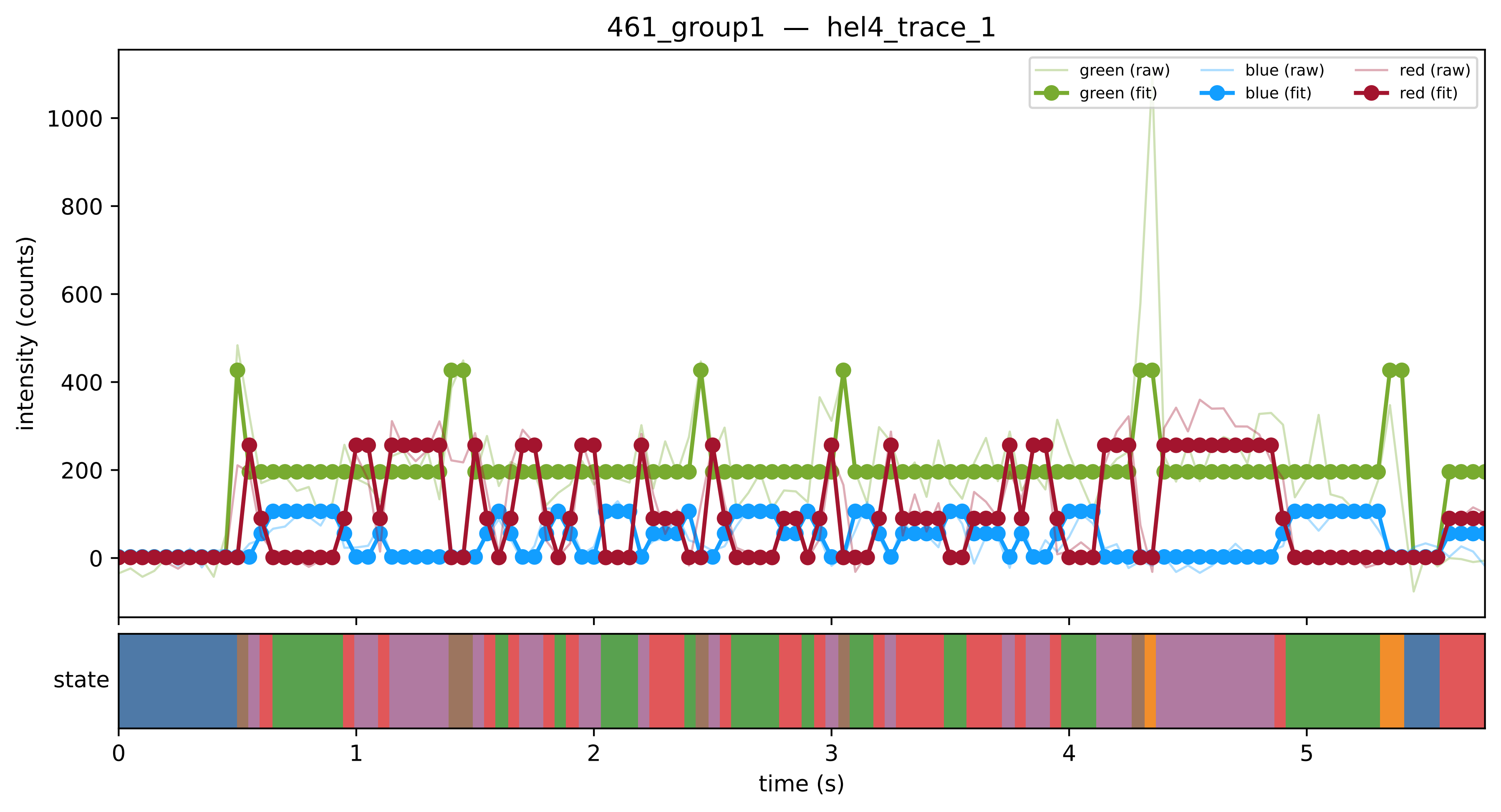

461_group1 — hel3_trace_2
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
500
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
time (s)



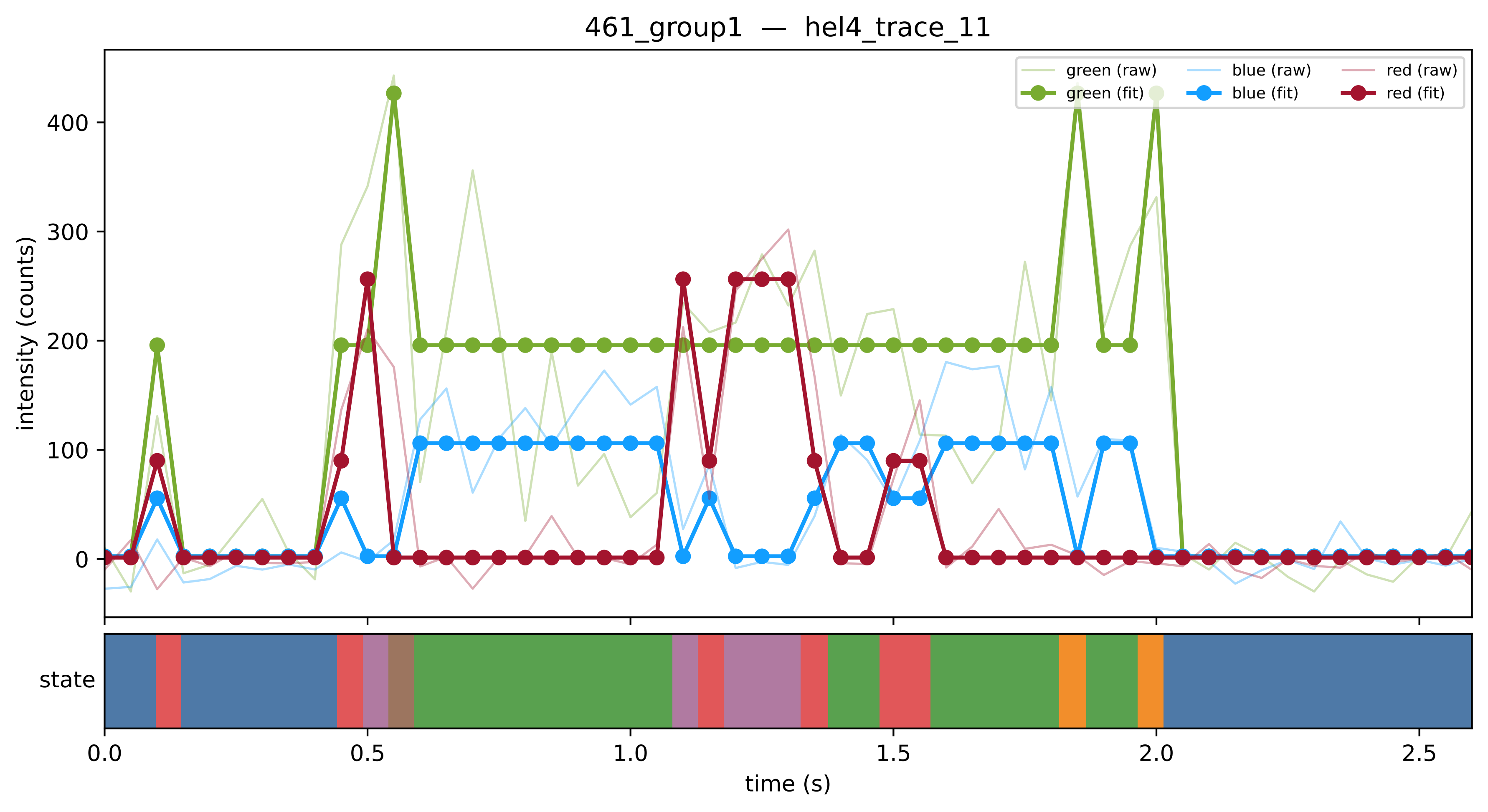
461_group1 — hel3_trace_29
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
300
250
200
150
100
50
0
−50
state
0.0
0.2
0.4
0.6
0.8
1.0
1.2
time (s)


461_group1 — hel3_trace_35
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
500
600
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)

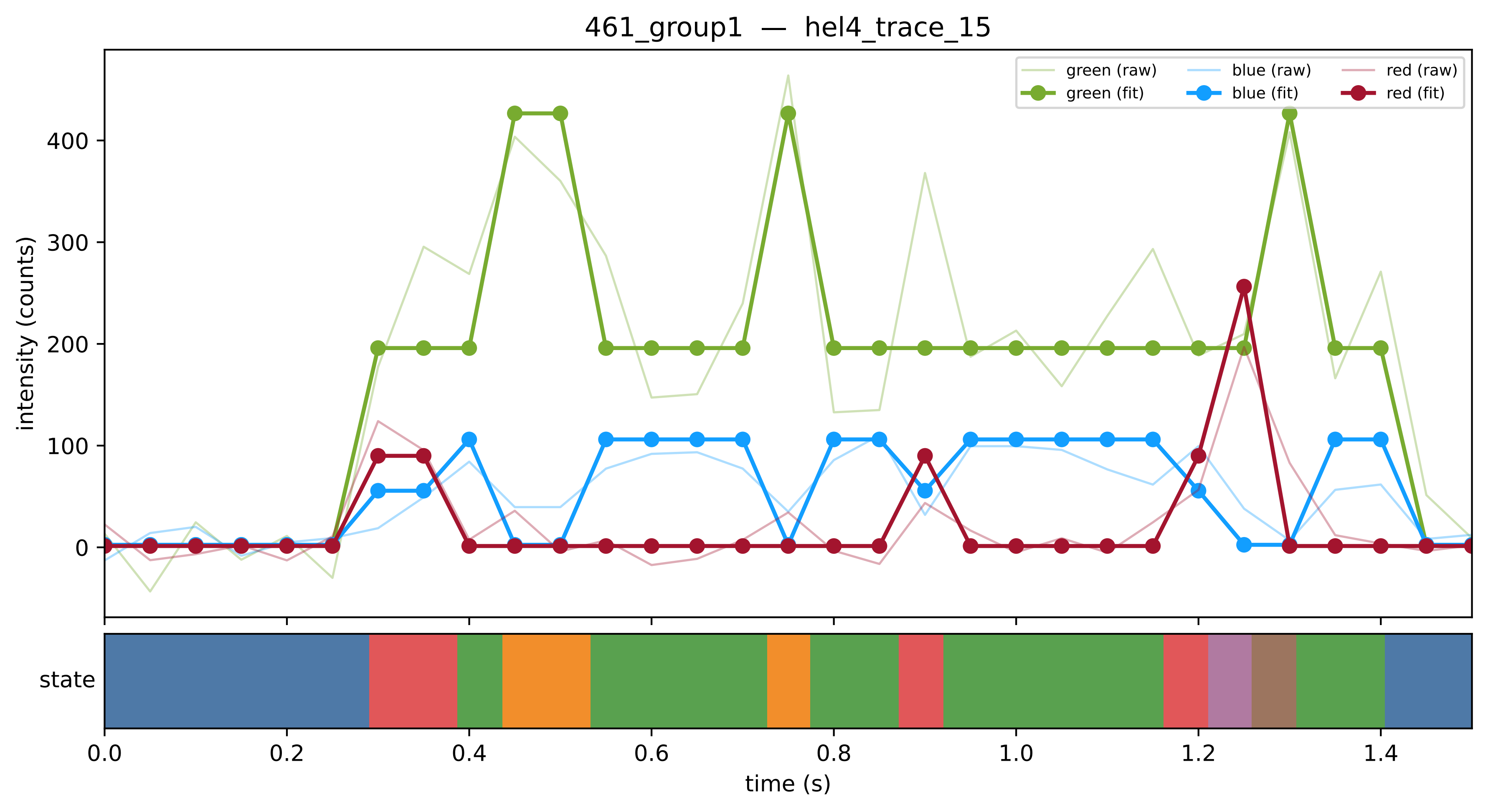

461_group1 — hel3_trace_5
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)



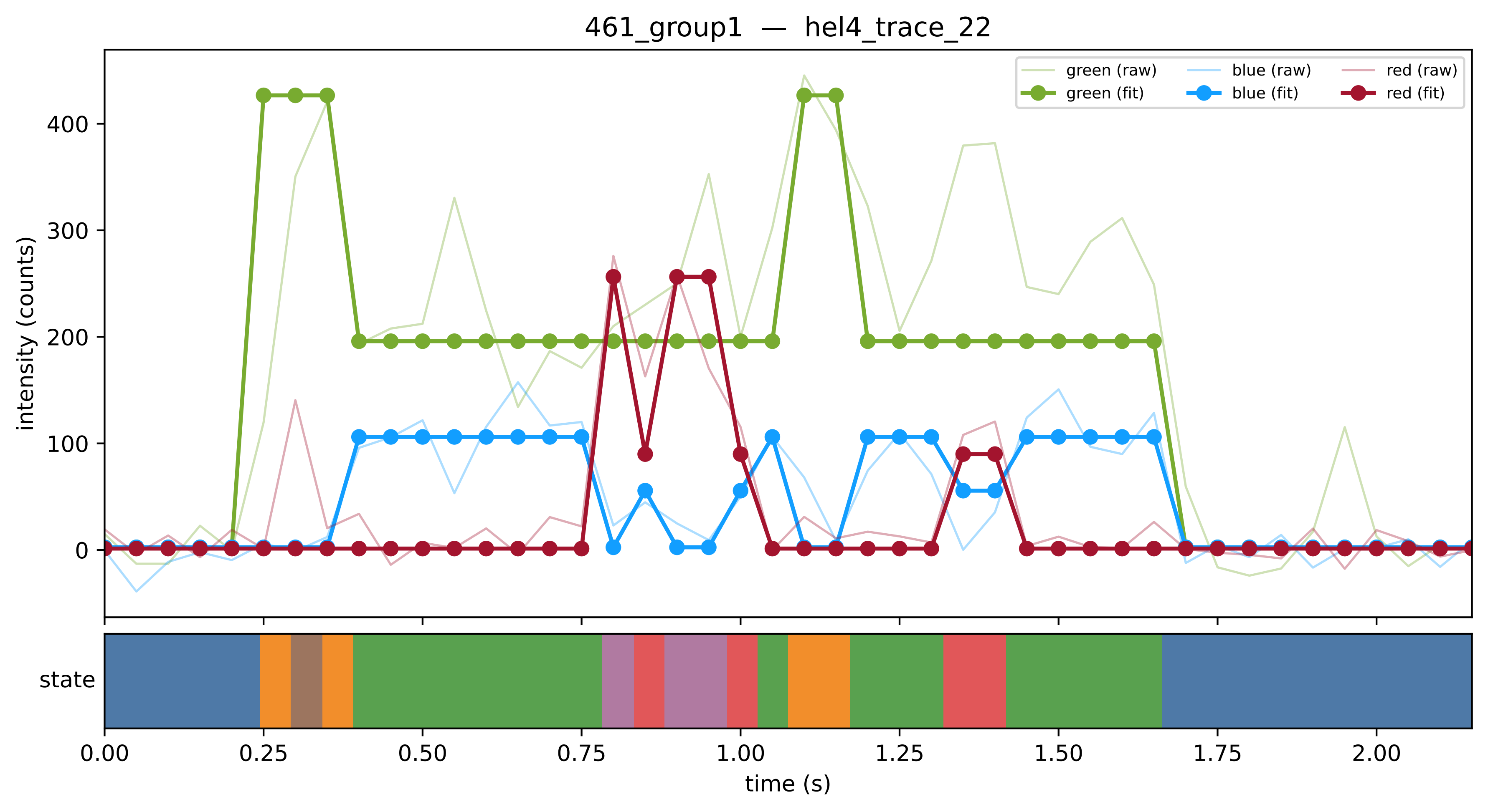
461_group1 — hel4_trace_1
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
1000
800
600
400
200
0
state
0
1
2
3
4
5
time (s)

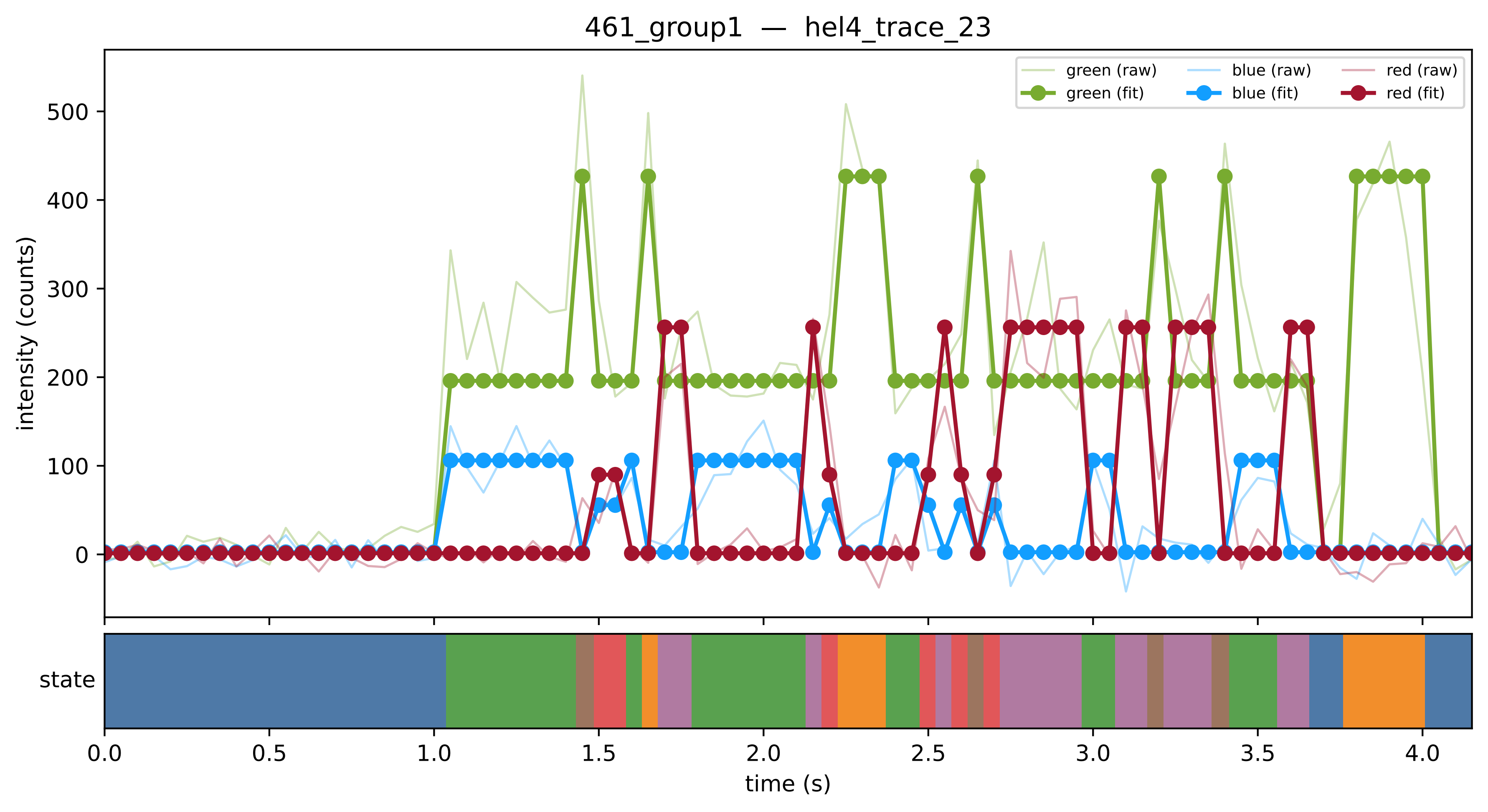


461_group1 — hel4_trace_11
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)

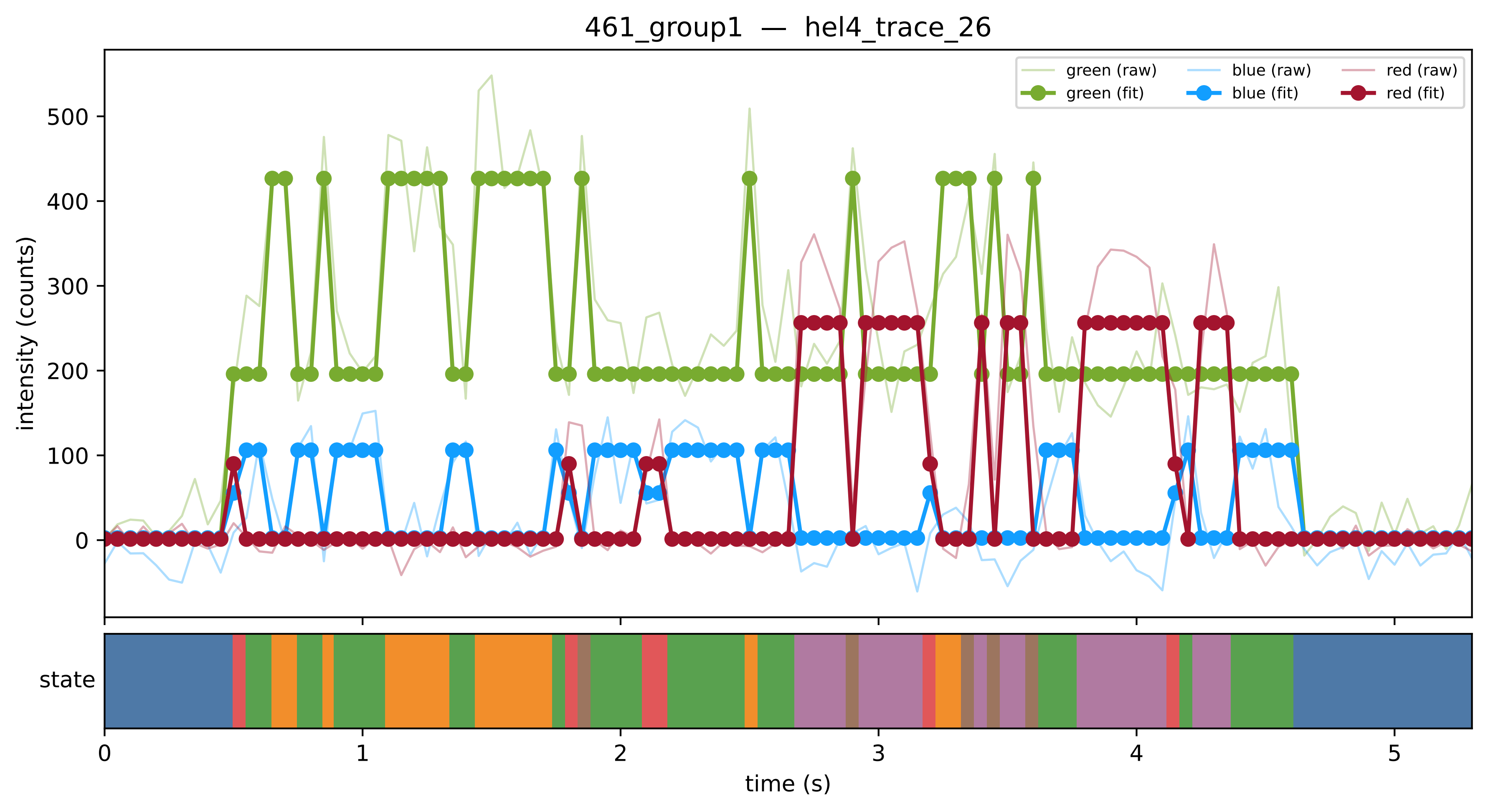


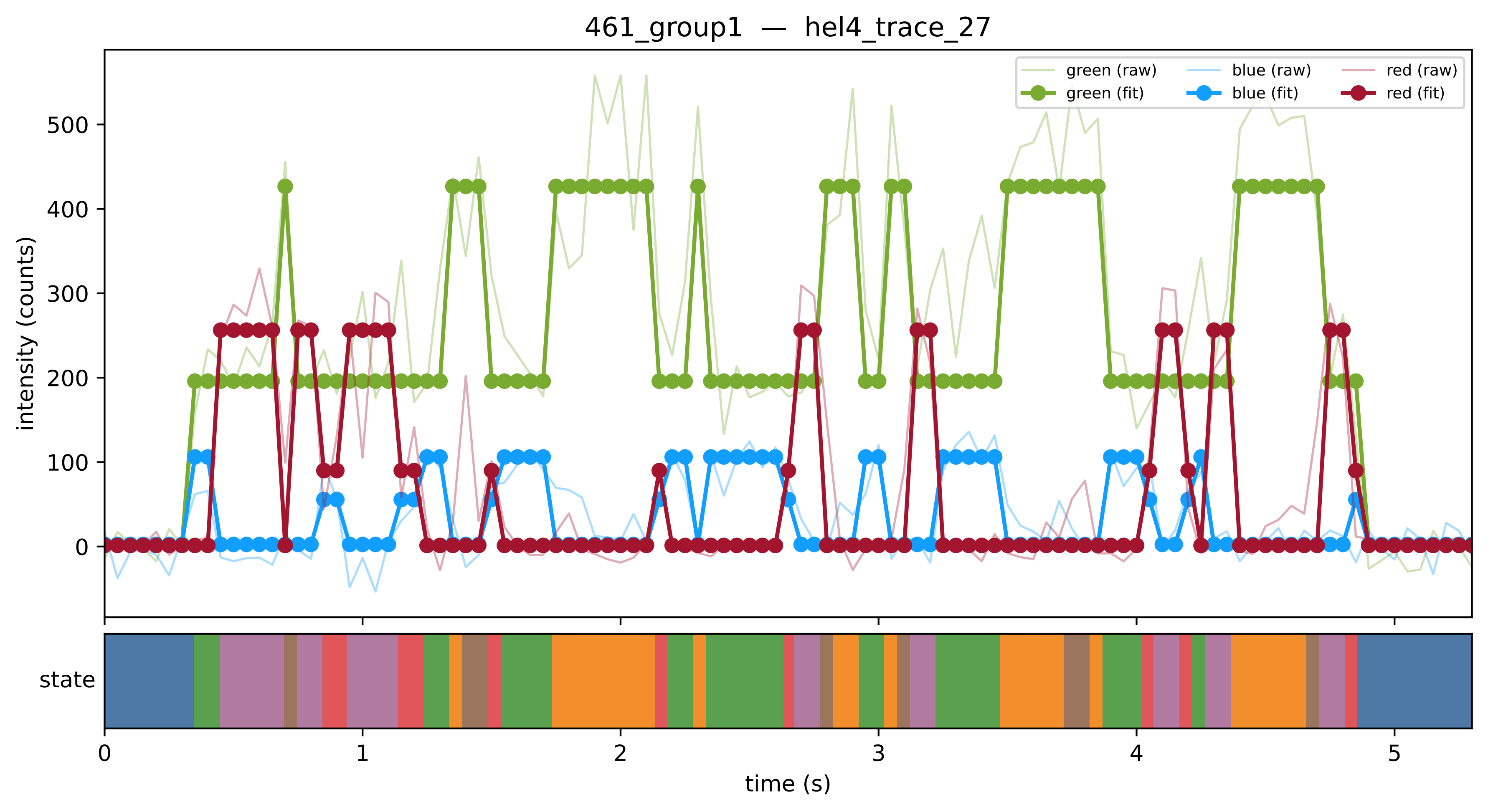
461_group1 — hel4_trace_15
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
time (s)

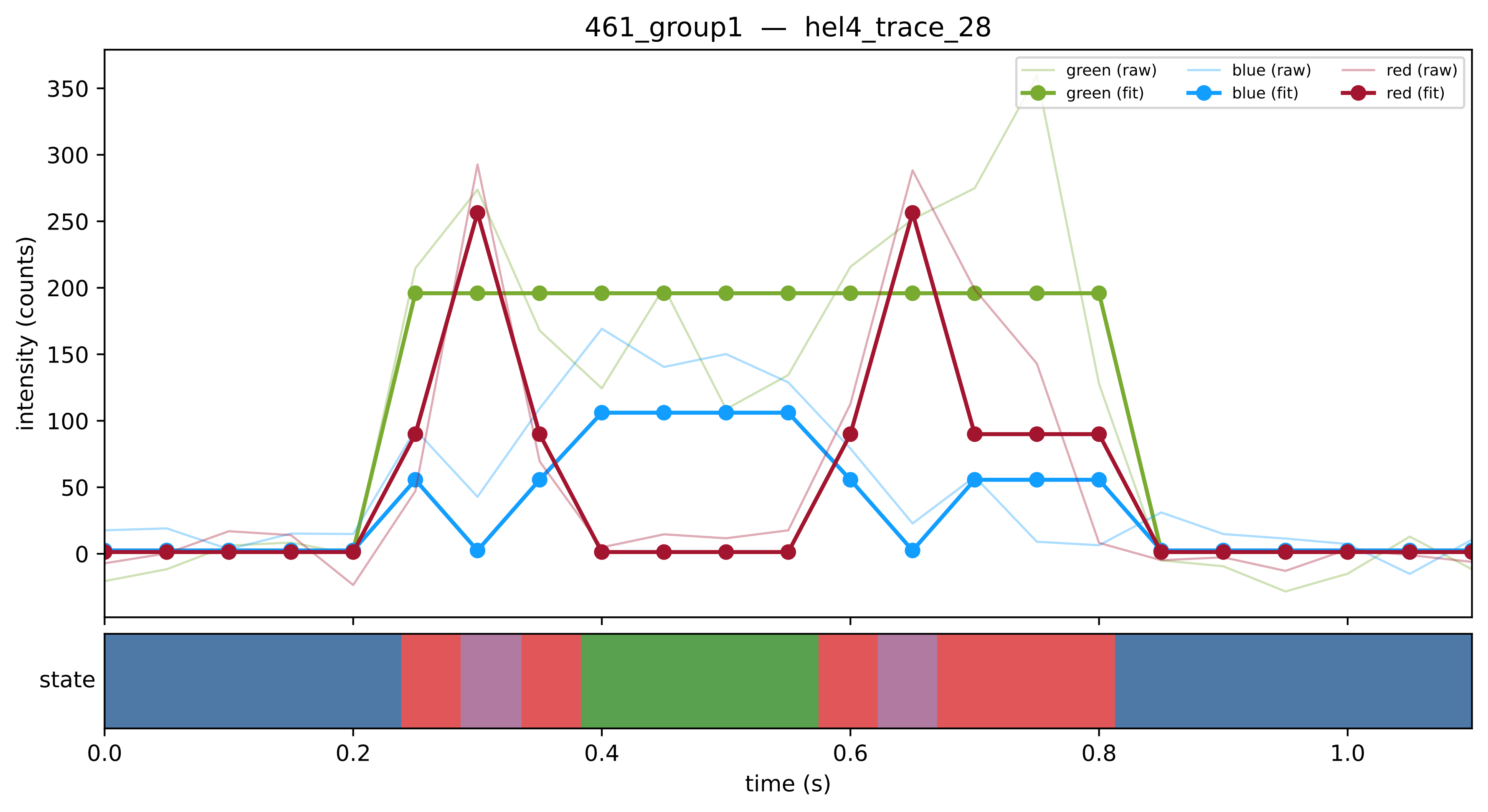

461_group1 — hel4_trace_22
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time (s)

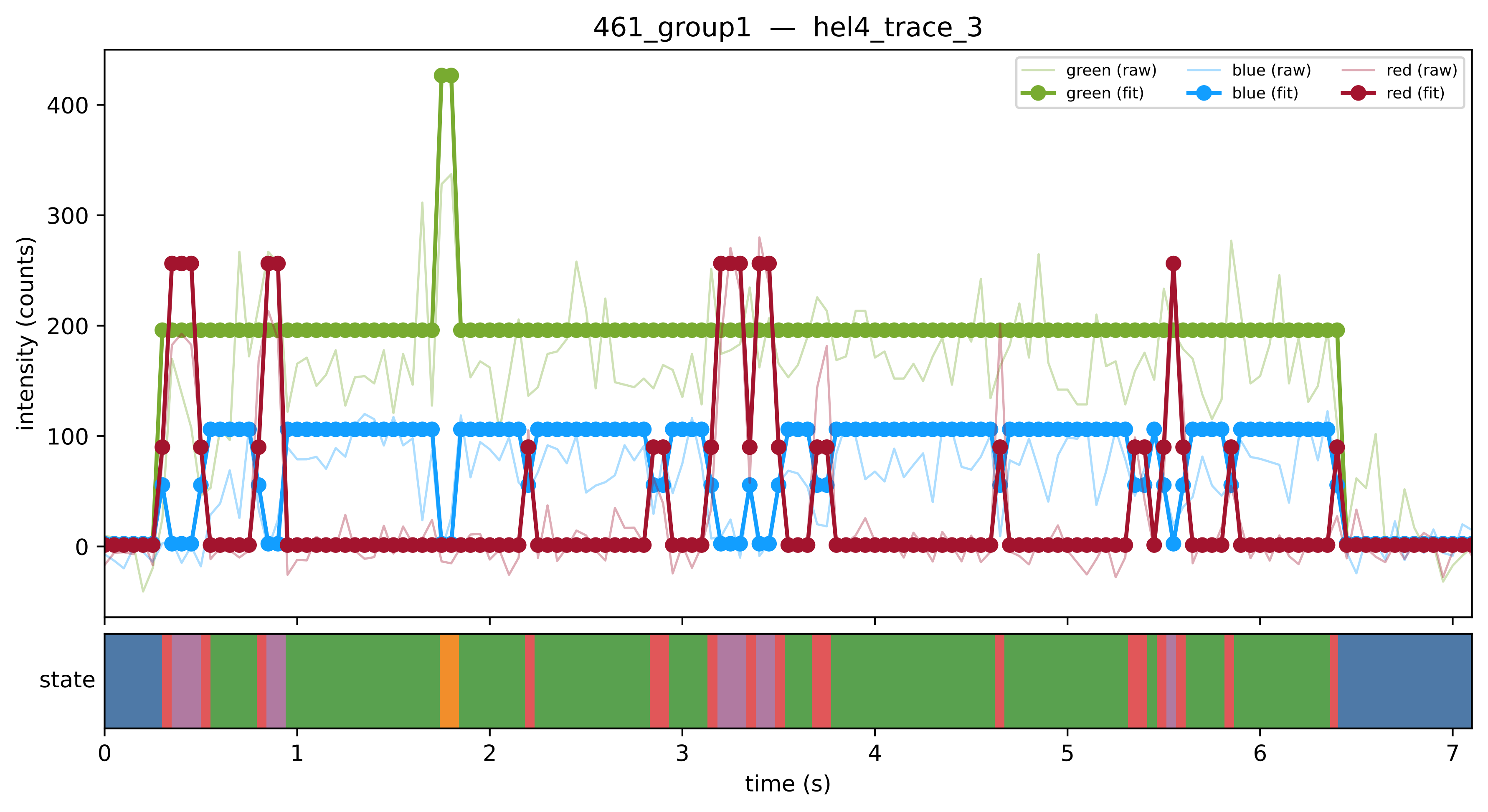


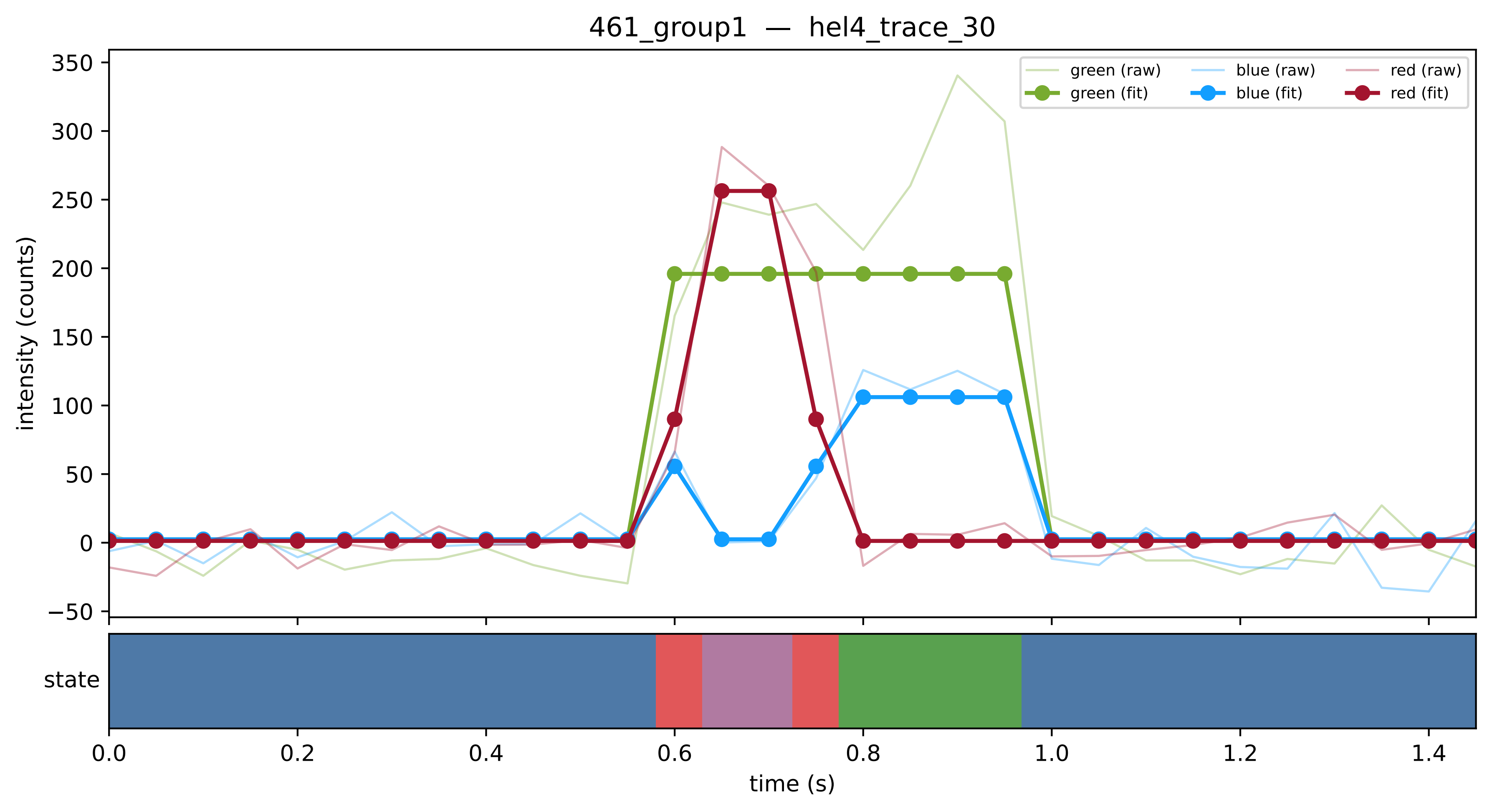
461_group1 — hel4_trace_23
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
500
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
time (s)

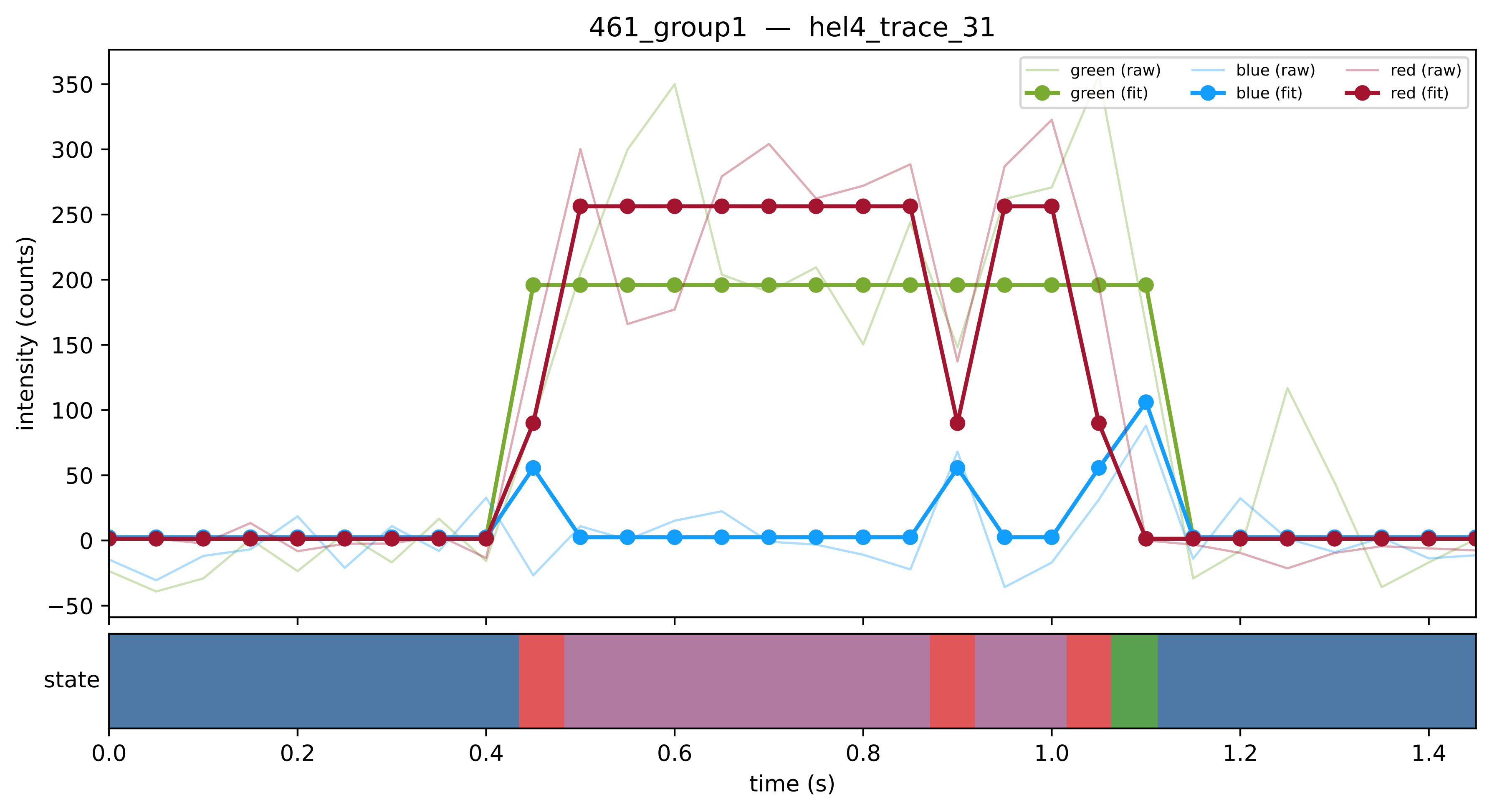

461_group1 — hel4_trace_26
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
0
100
200
300
400
500
state
time (s)
0
1
2
3
4
5


461_group1 — hel4_trace_27
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
500
state
0
1
2
3
4
5
time (s)

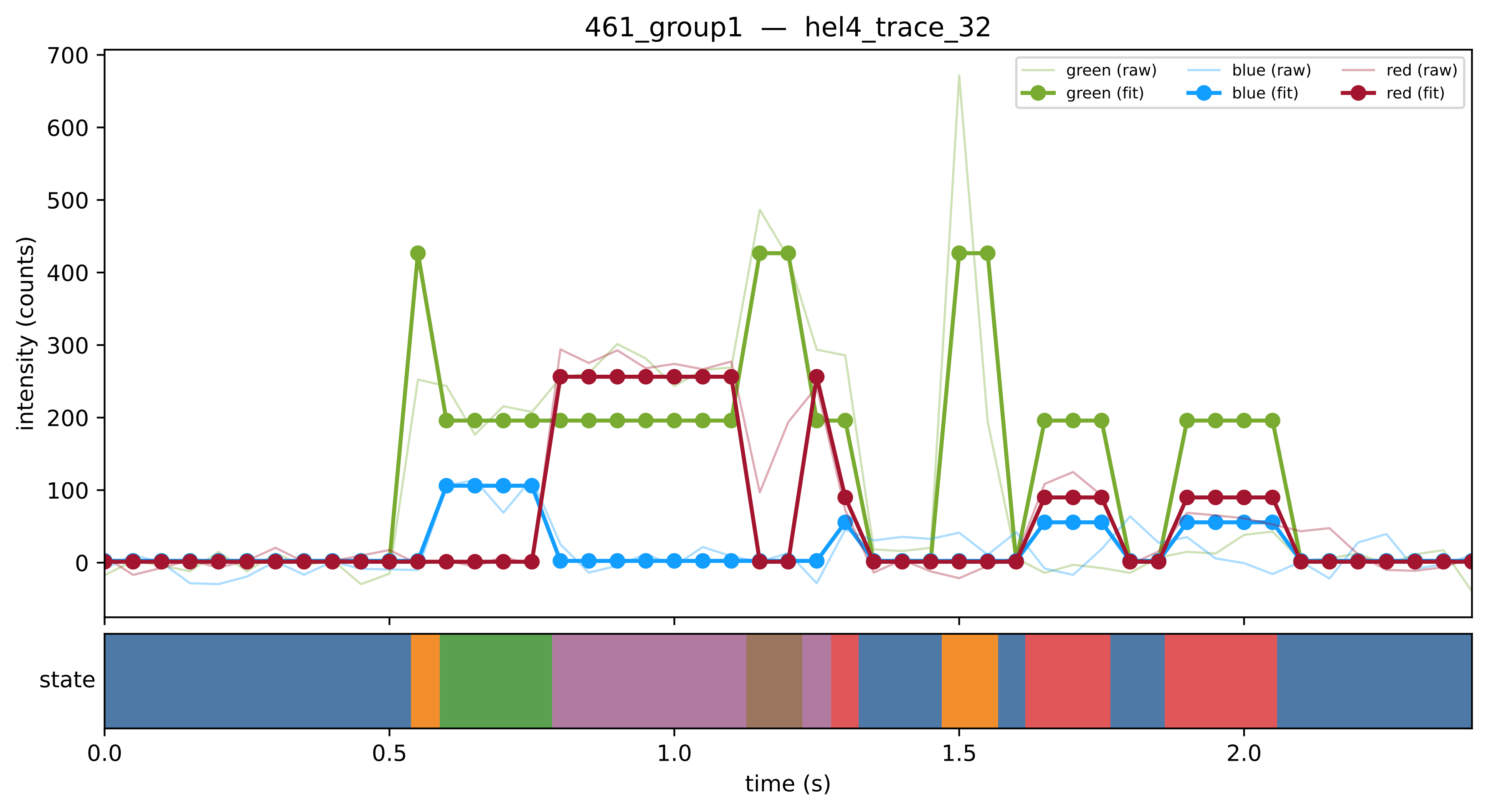

461_group1 — hel4_trace_28
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
350
300
250
200
150
100
50
0
state
0.0
0.2
0.4
0.6
0.8
1.0
time (s)

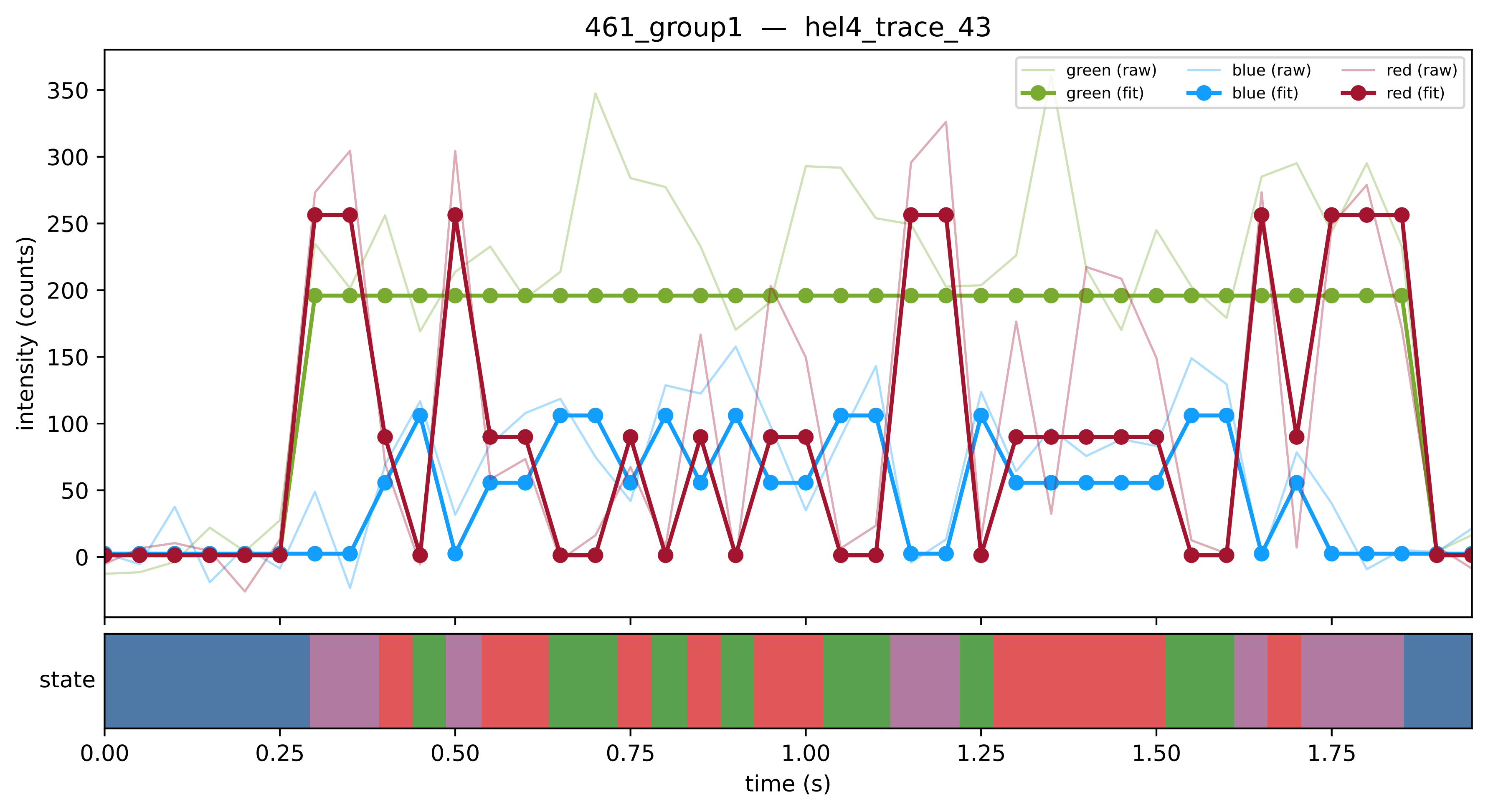

461_group1 — hel4_trace_3
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0
1
2
3
4
5
6
7
time (s)

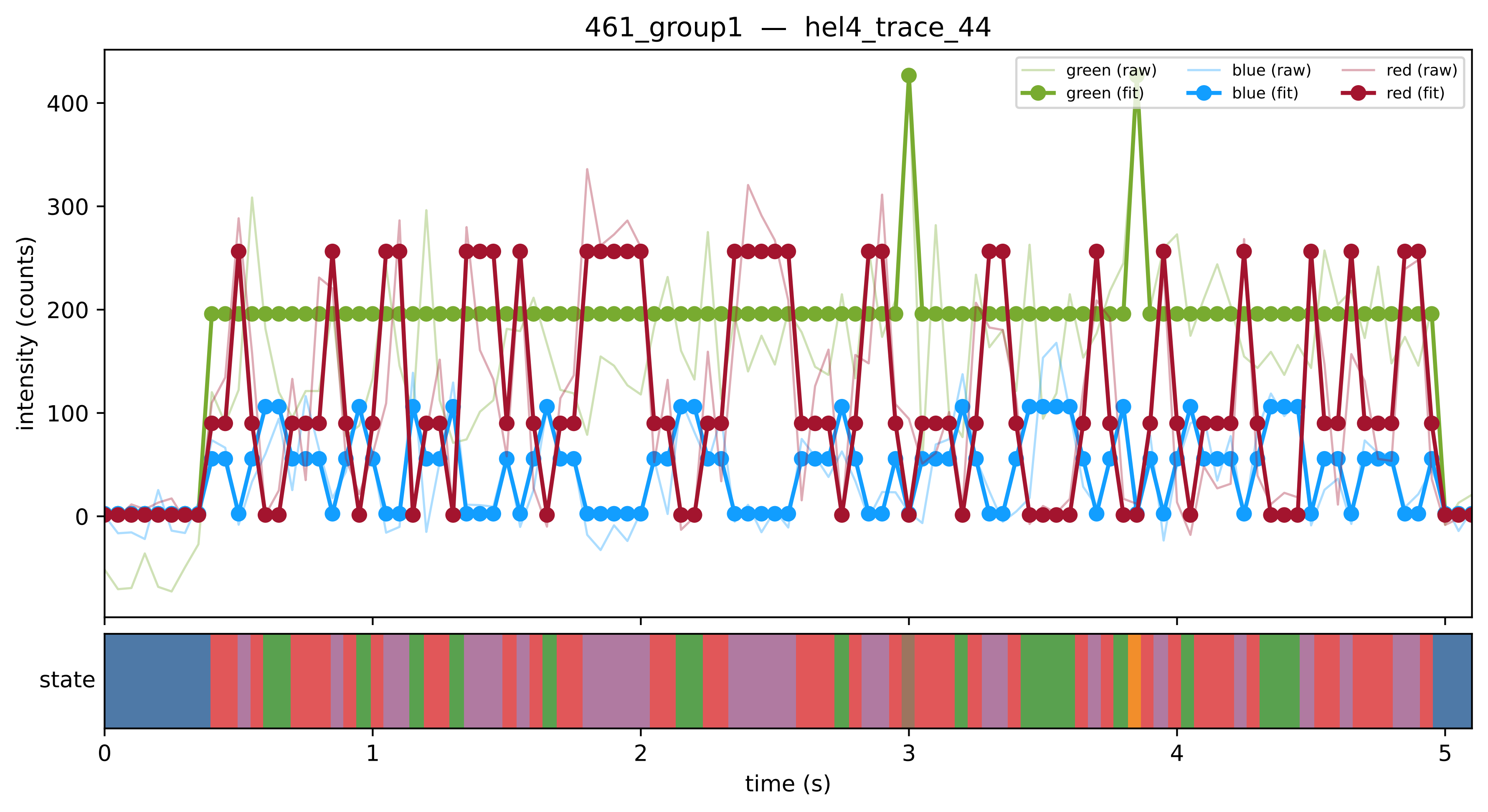

461_group1 — hel4_trace_30
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
350
300
250
200
150
100
50
0
−50
state
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
time (s)

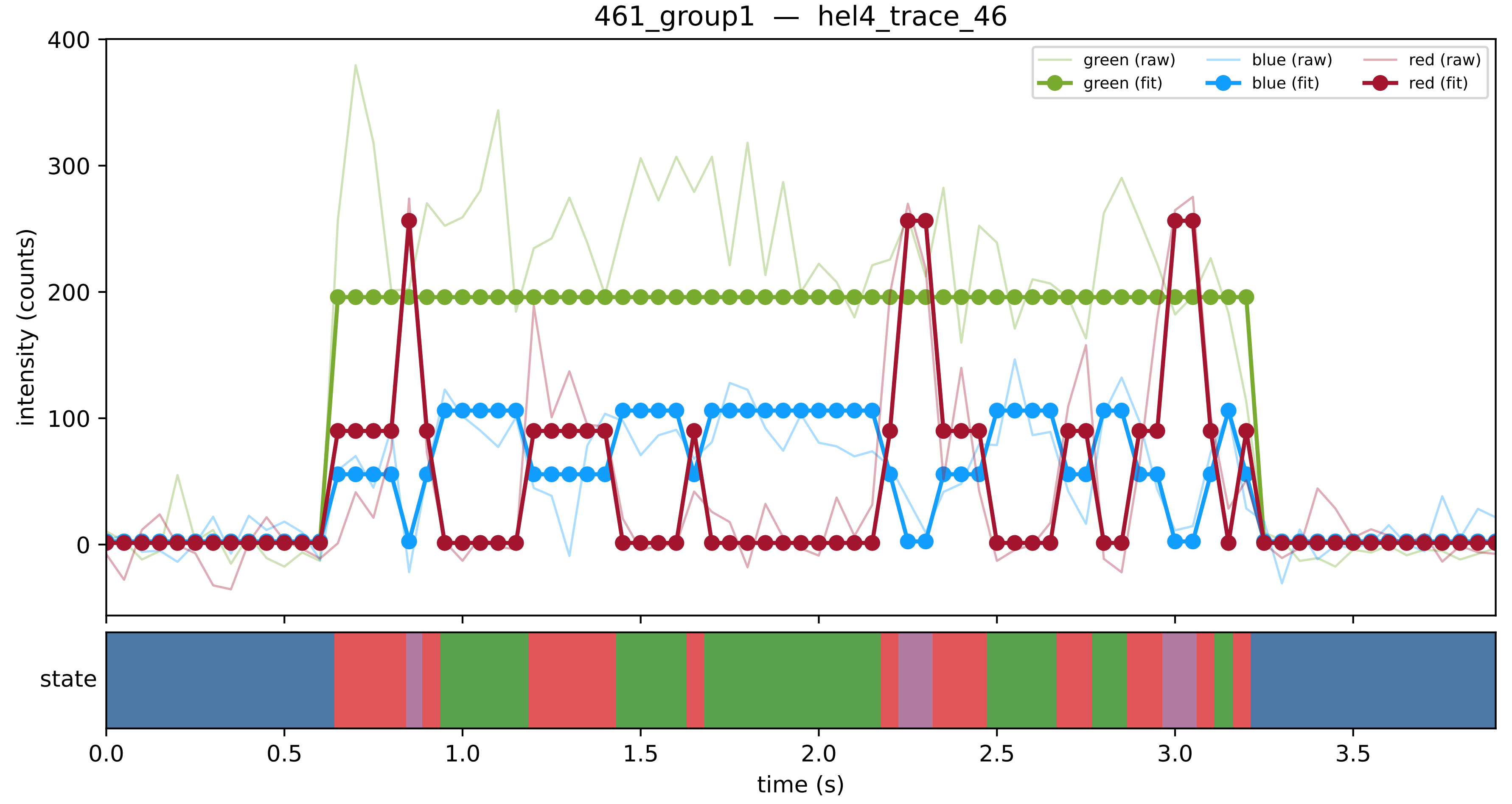


461_group1 — hel4_trace_31
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
350
300
250
200
150
100
50
0
−50
state
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
time (s)

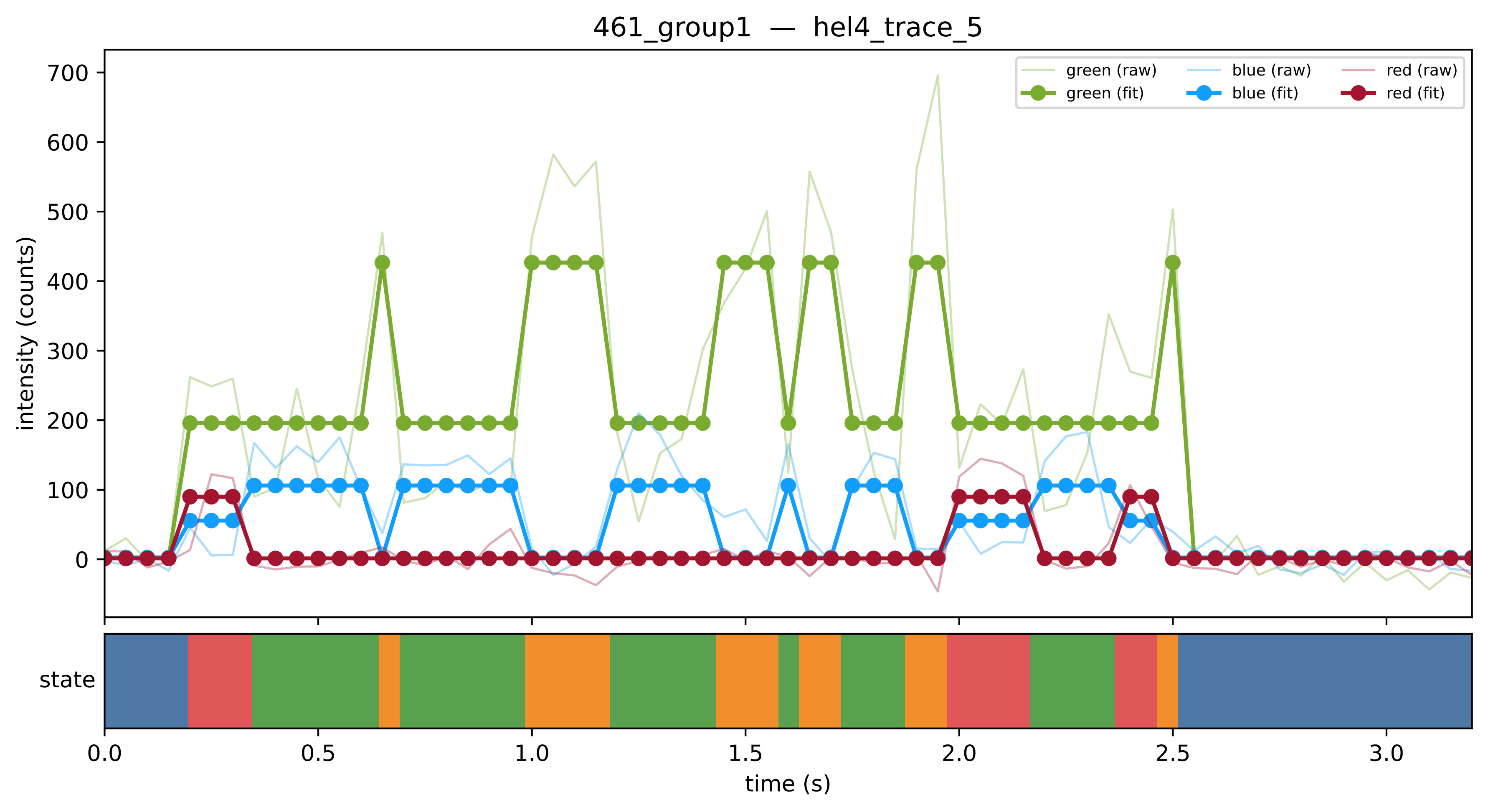

461_group1 — hel4_trace_32
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
100
200
300
400
500
600
700
0.0
0.5
1.0
1.5
2.0

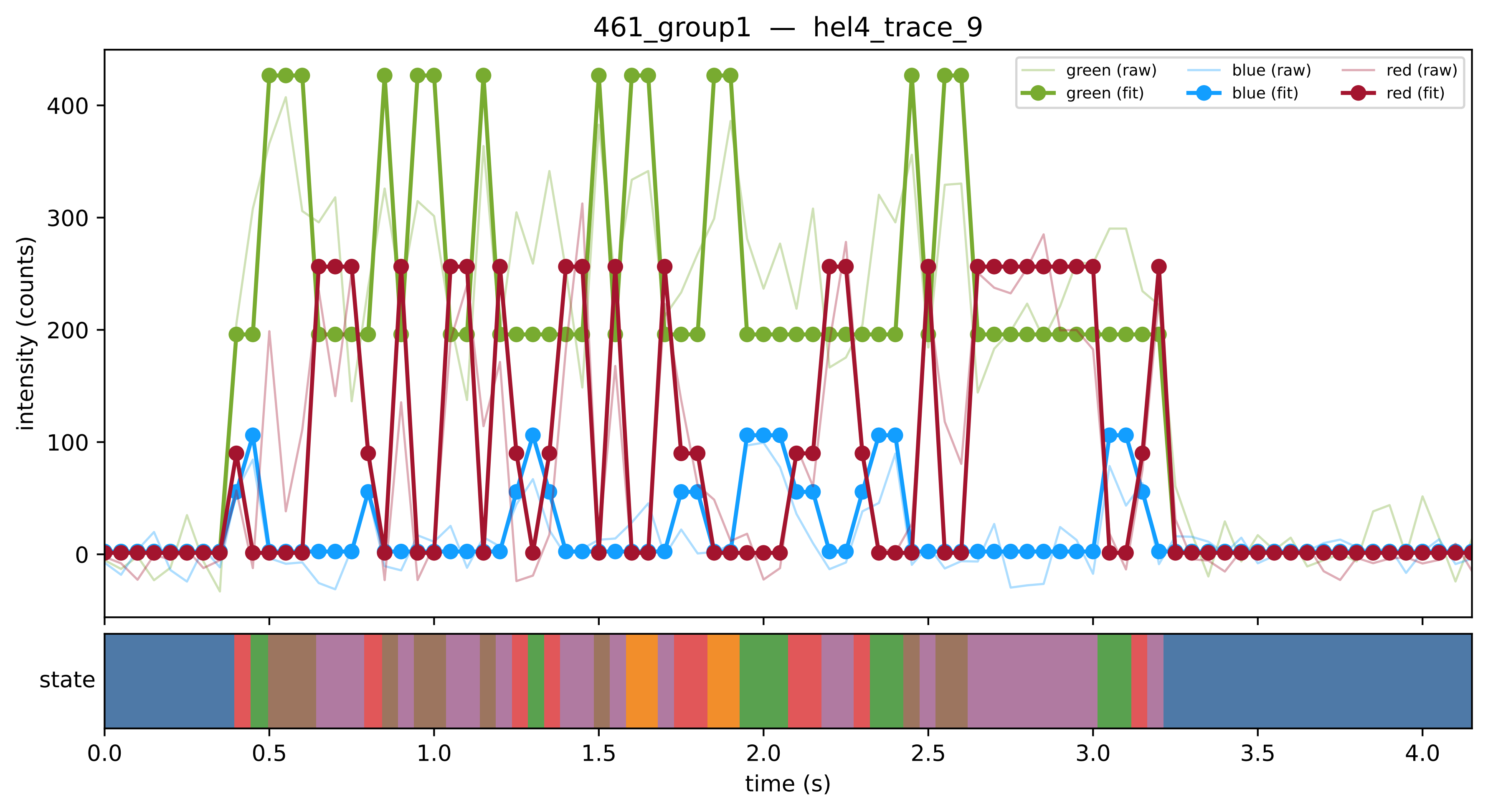

461_group1 — hel4_trace_43
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
50
100
150
200
250
300
350
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75


461_group1 — hel4_trace_44
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
0
100
200
300
400
state
0
1
2
3
4
5
time (s)

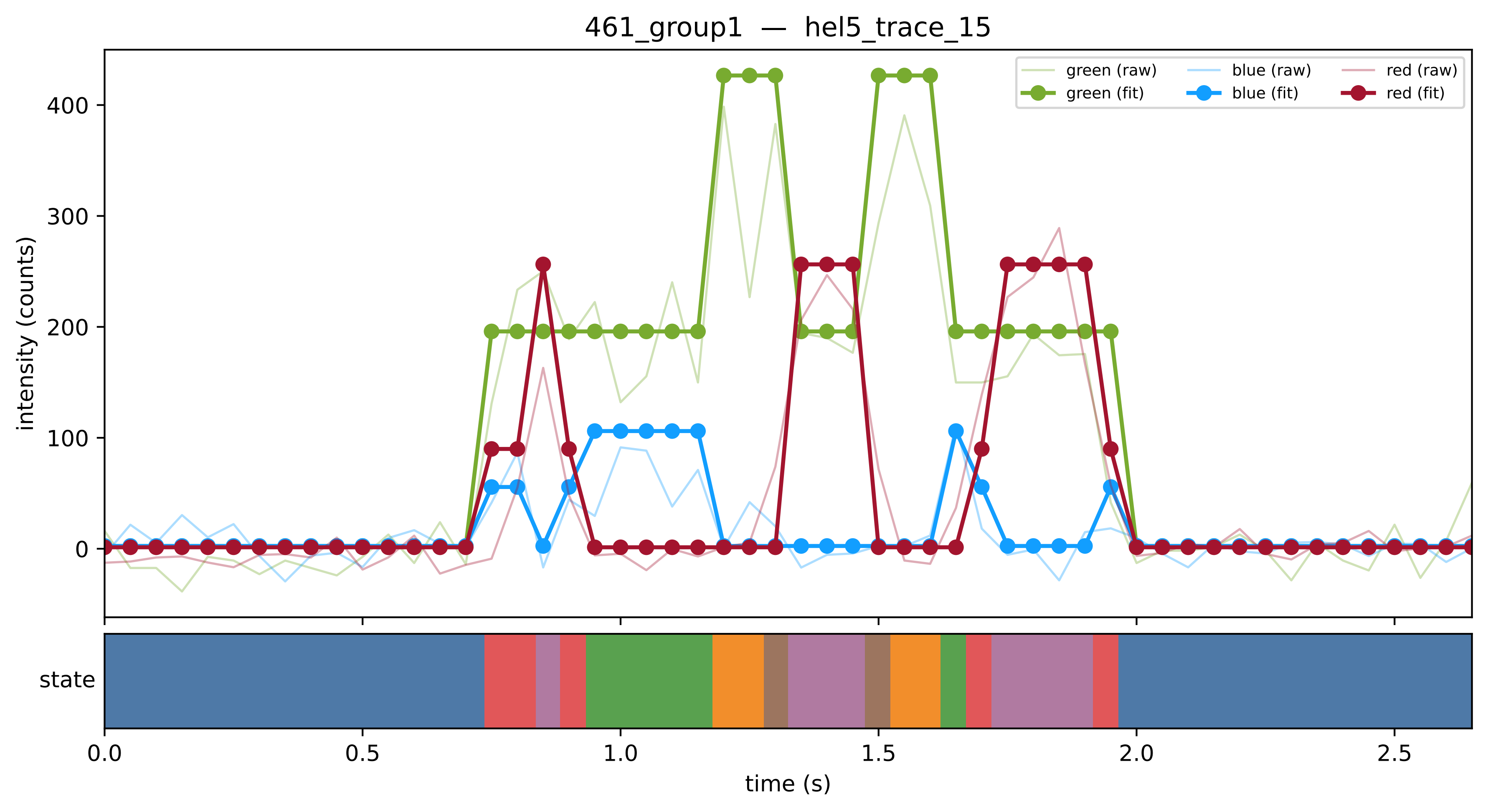


461_group1 — hel4_trace_46
intensity (counts)
400
300
200
100
0
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
time (s)

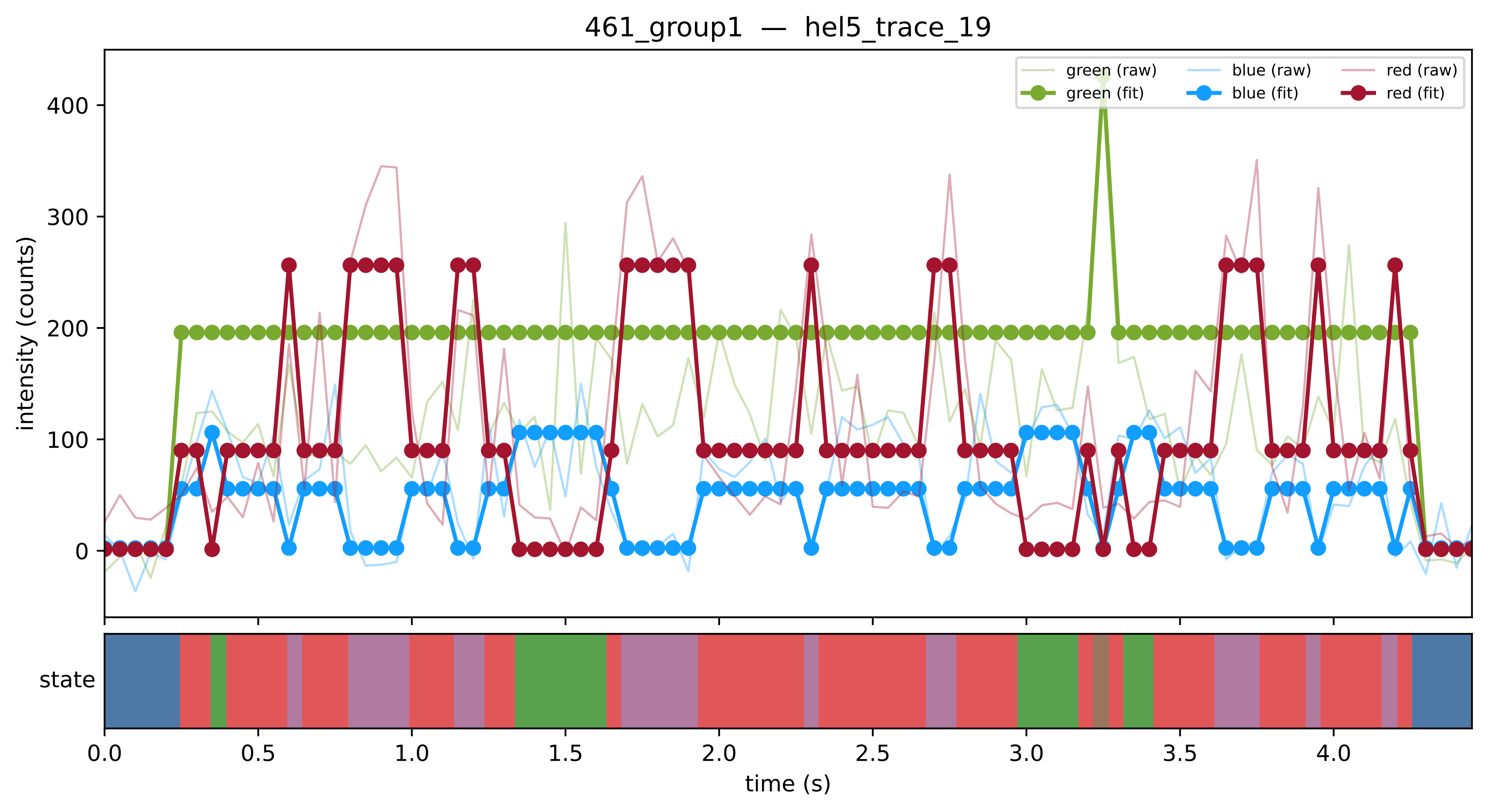

461_group1 — hel4_trace_5
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
700
600
500
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
time (s)

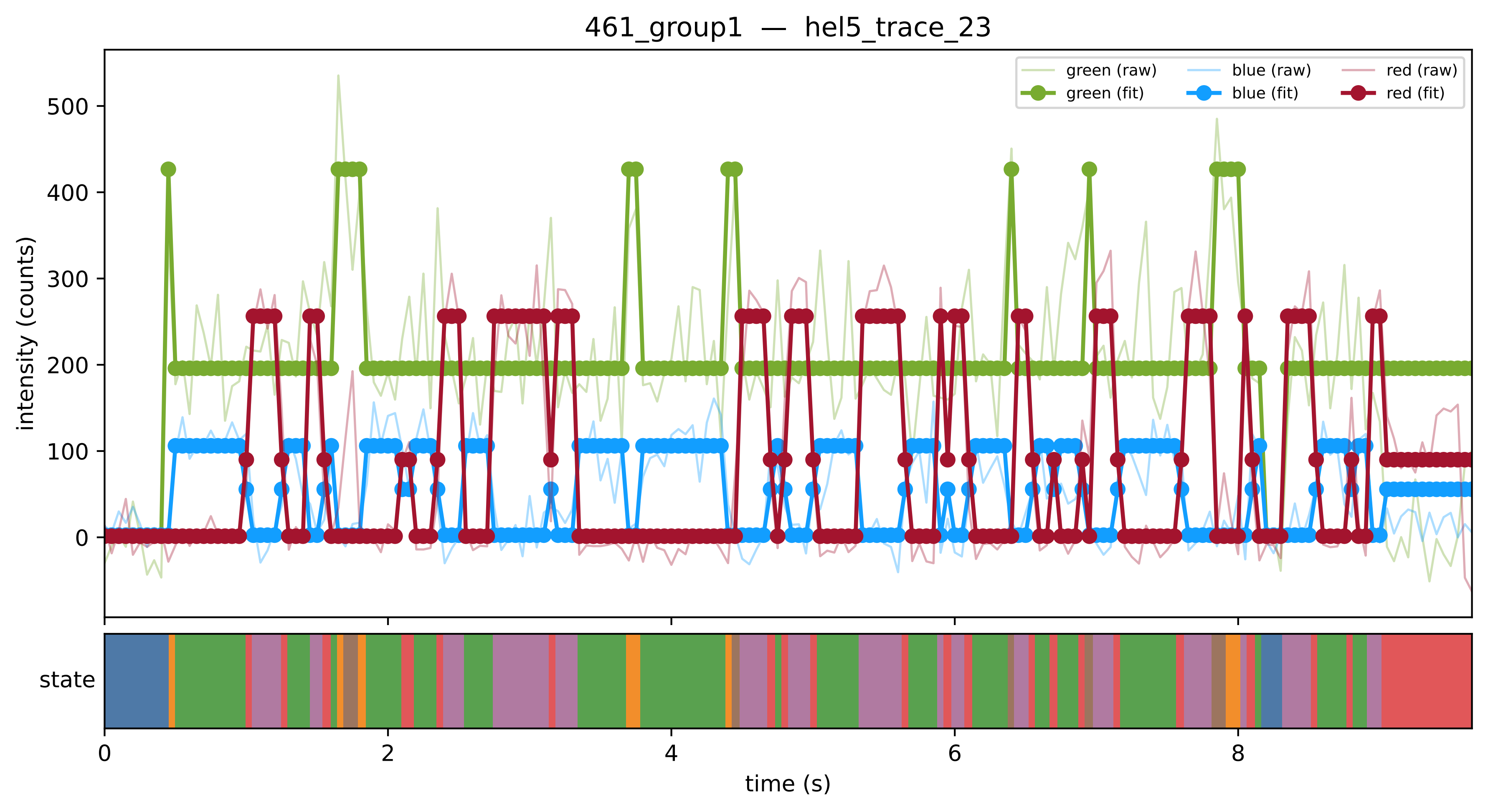


461_group1 — hel4_trace_9
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
100
200
300
400
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

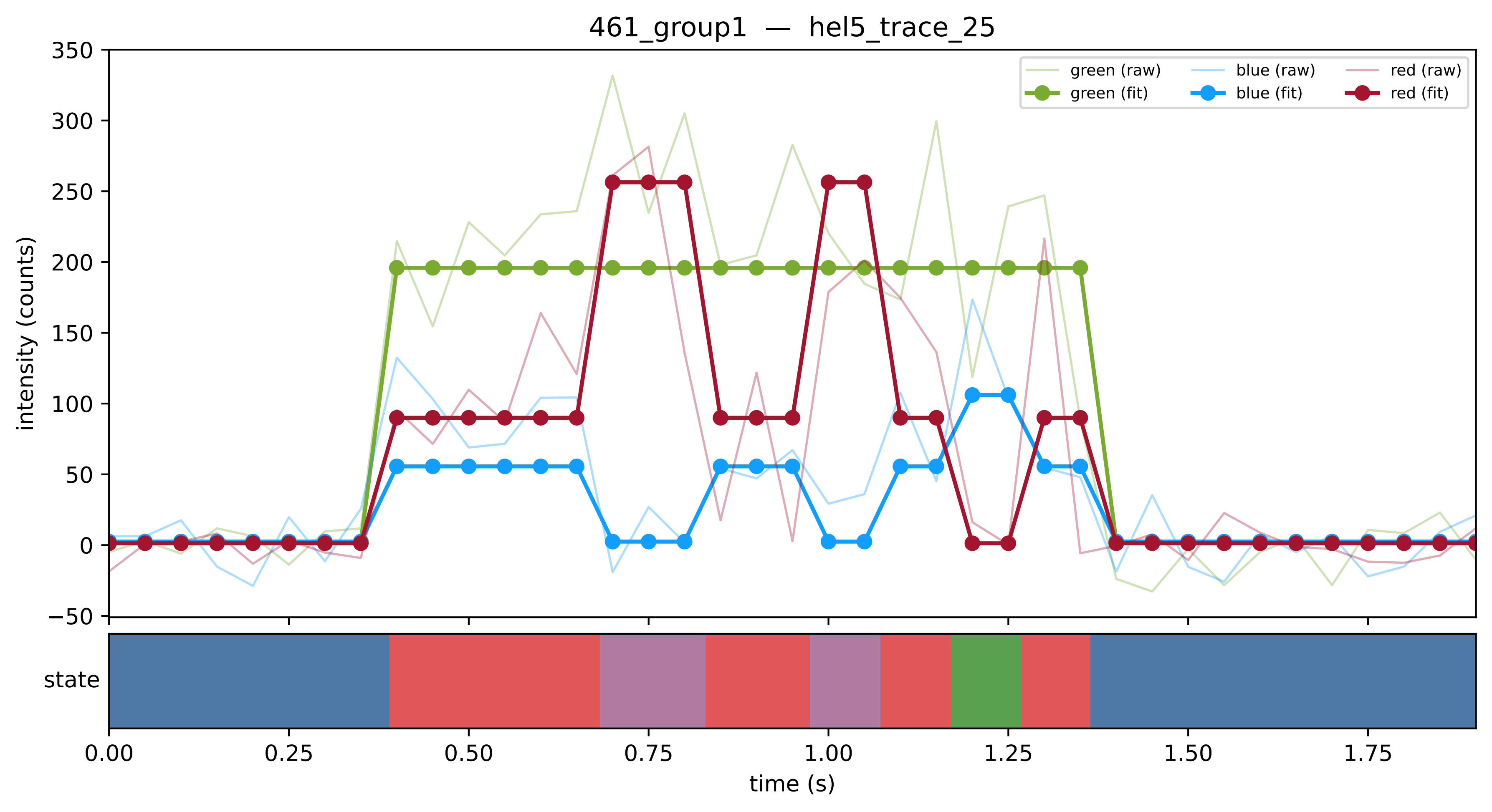


461_group1 — hel5_trace_15
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)

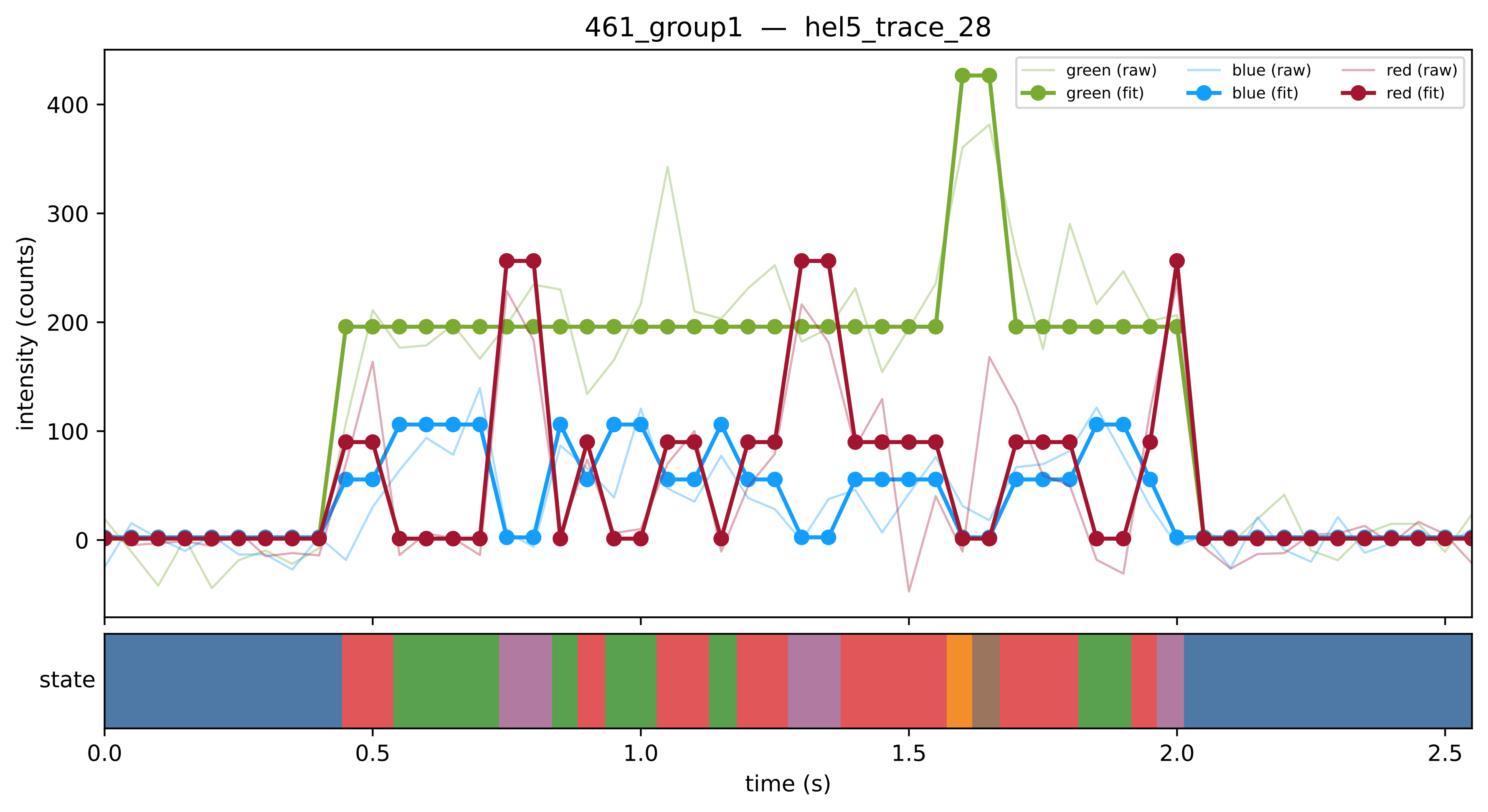

461_group1 — hel5_trace_19
intensity (counts)
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
state
time (s)

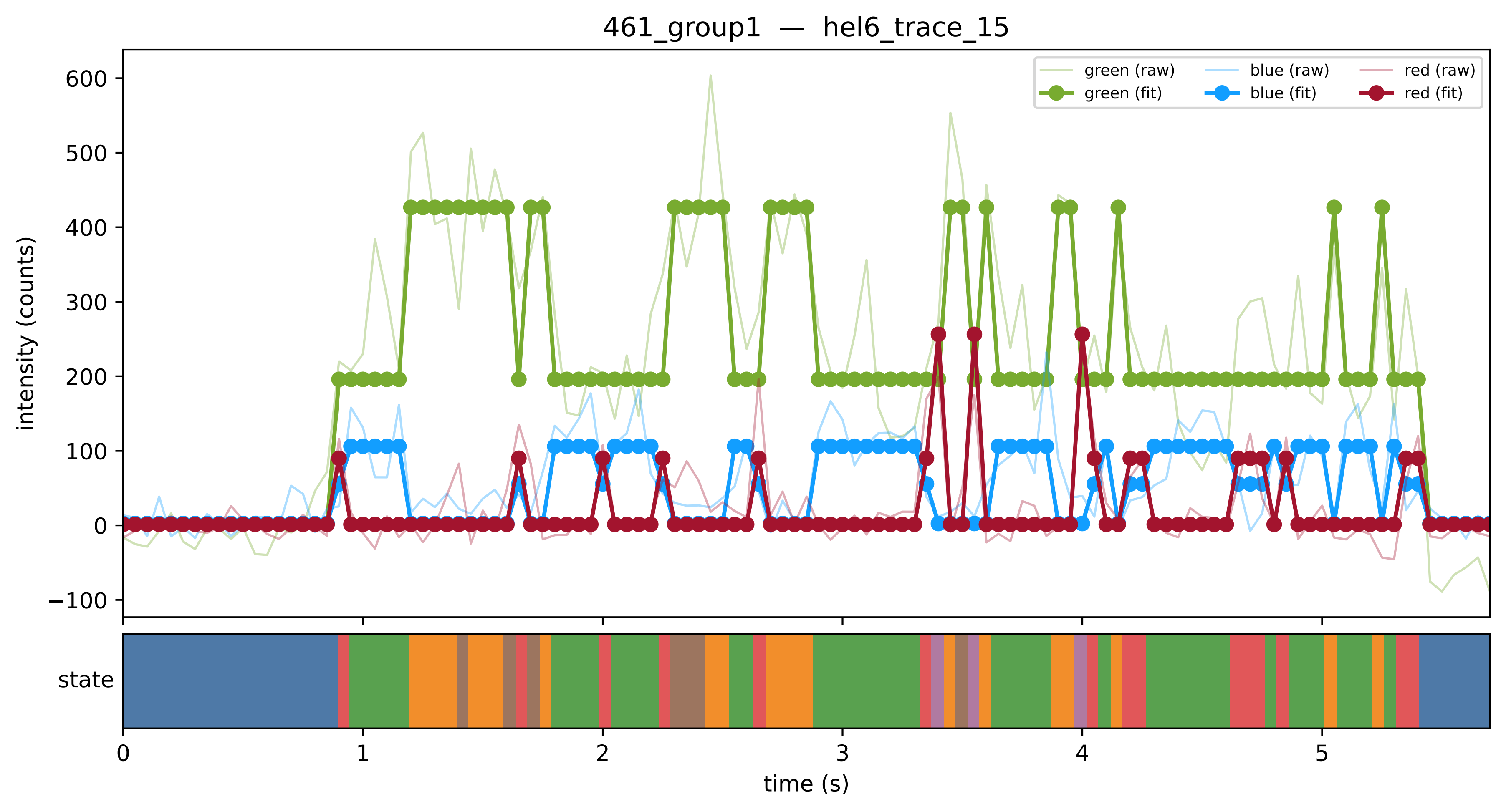


461_group1 — hel5_trace_23
intensity (counts)
500
400
300
200
100
0
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0
2
4
6
8
time (s)

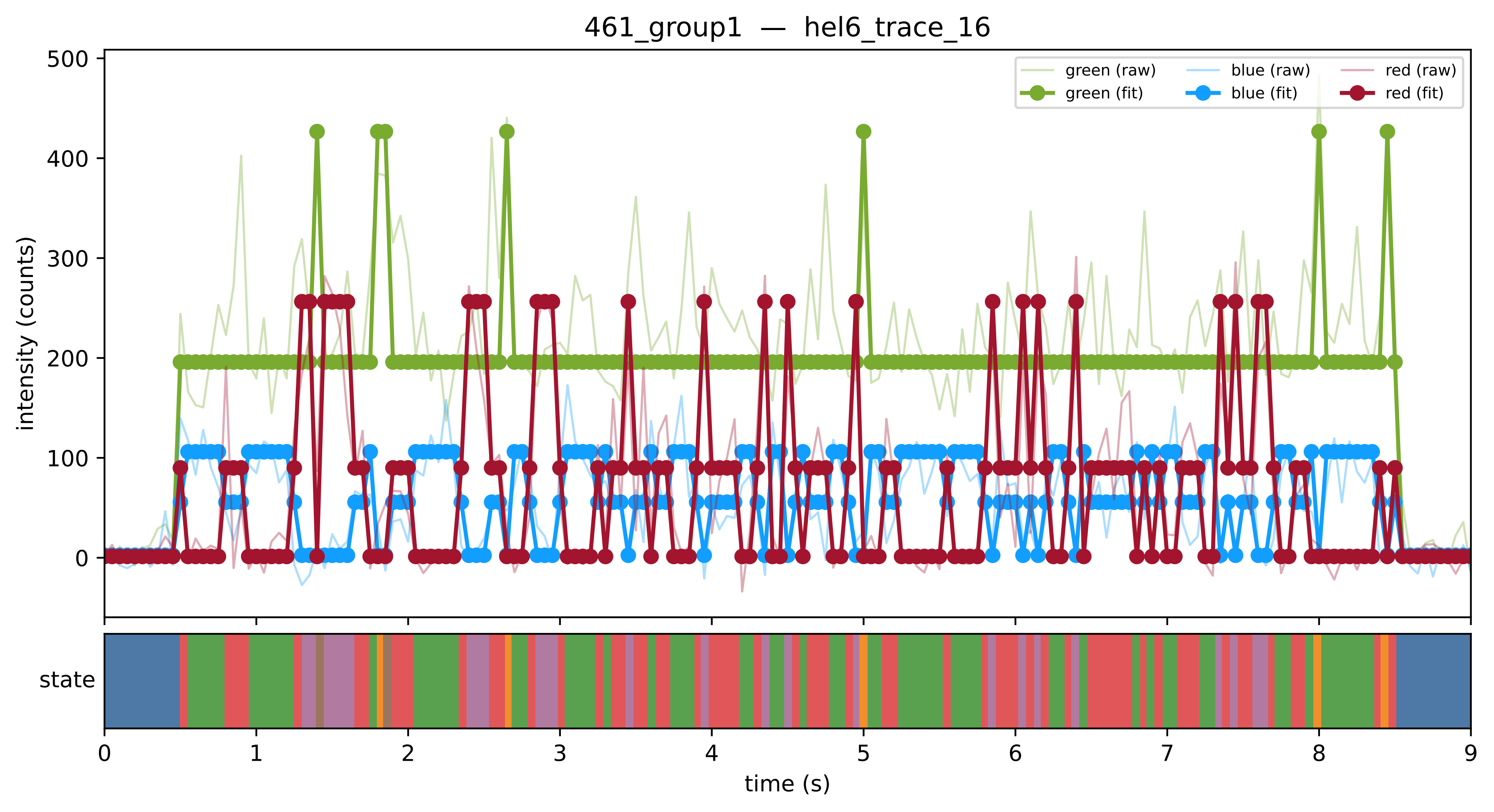


461_group1 — hel5_trace_25
intensity (counts)
350
300
250
200
150
100
50
0
−50
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
time (s)

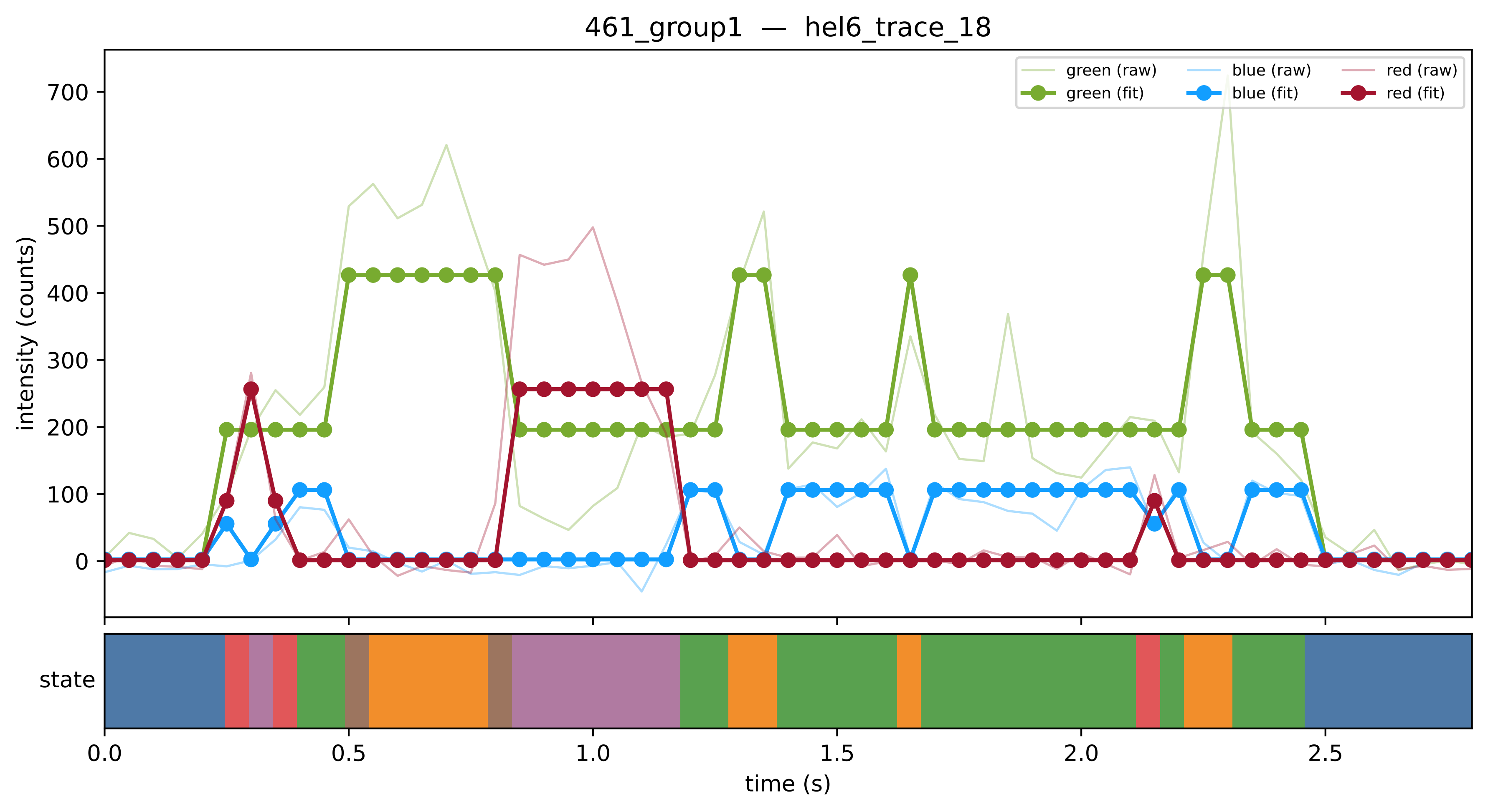

461_group1 — hel5_trace_28
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)



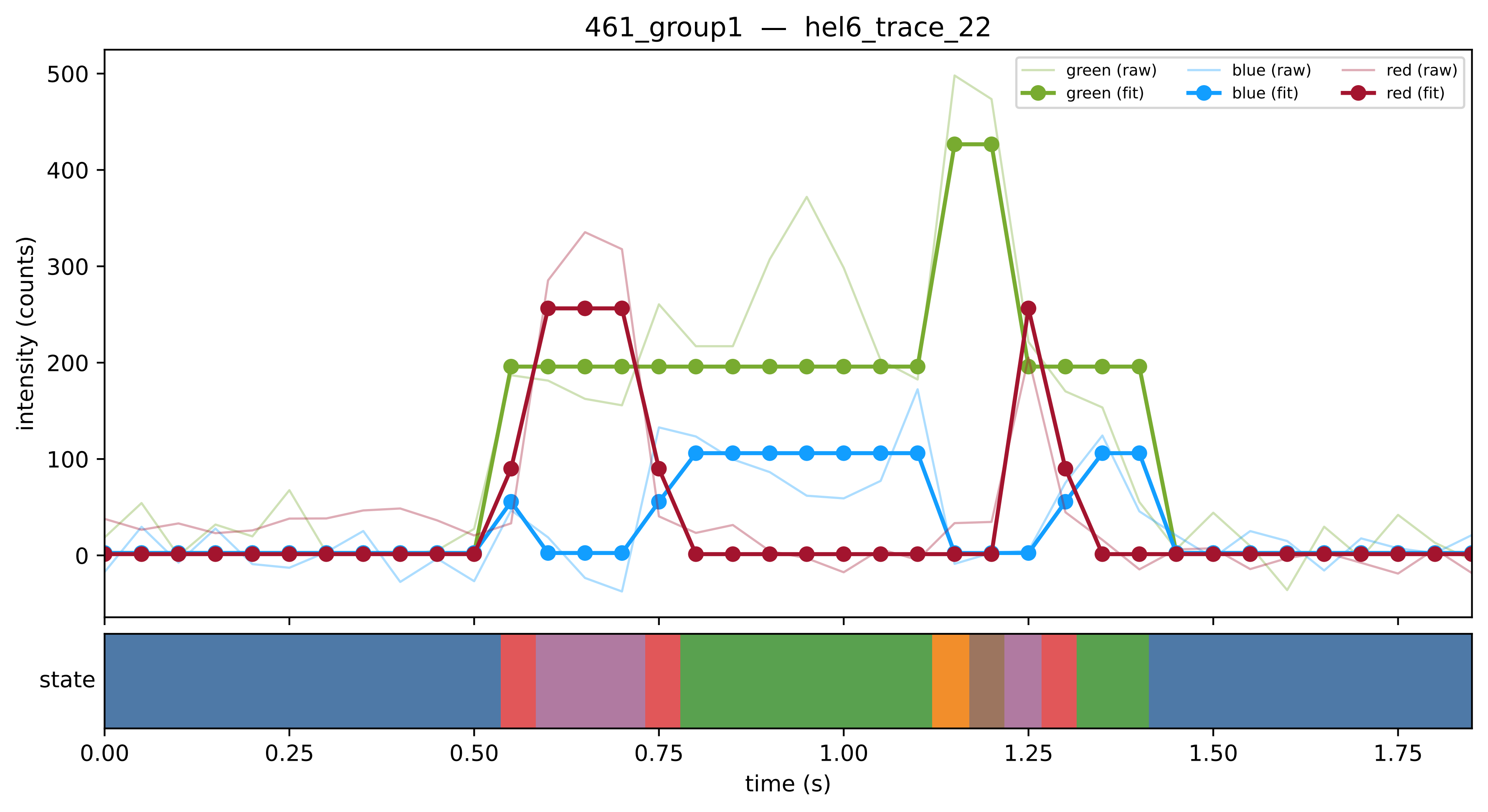
461_group1 — hel6_trace_15
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
600
500
400
300
200
100
0
−100
state
0
1
2
3
4
5
time (s)



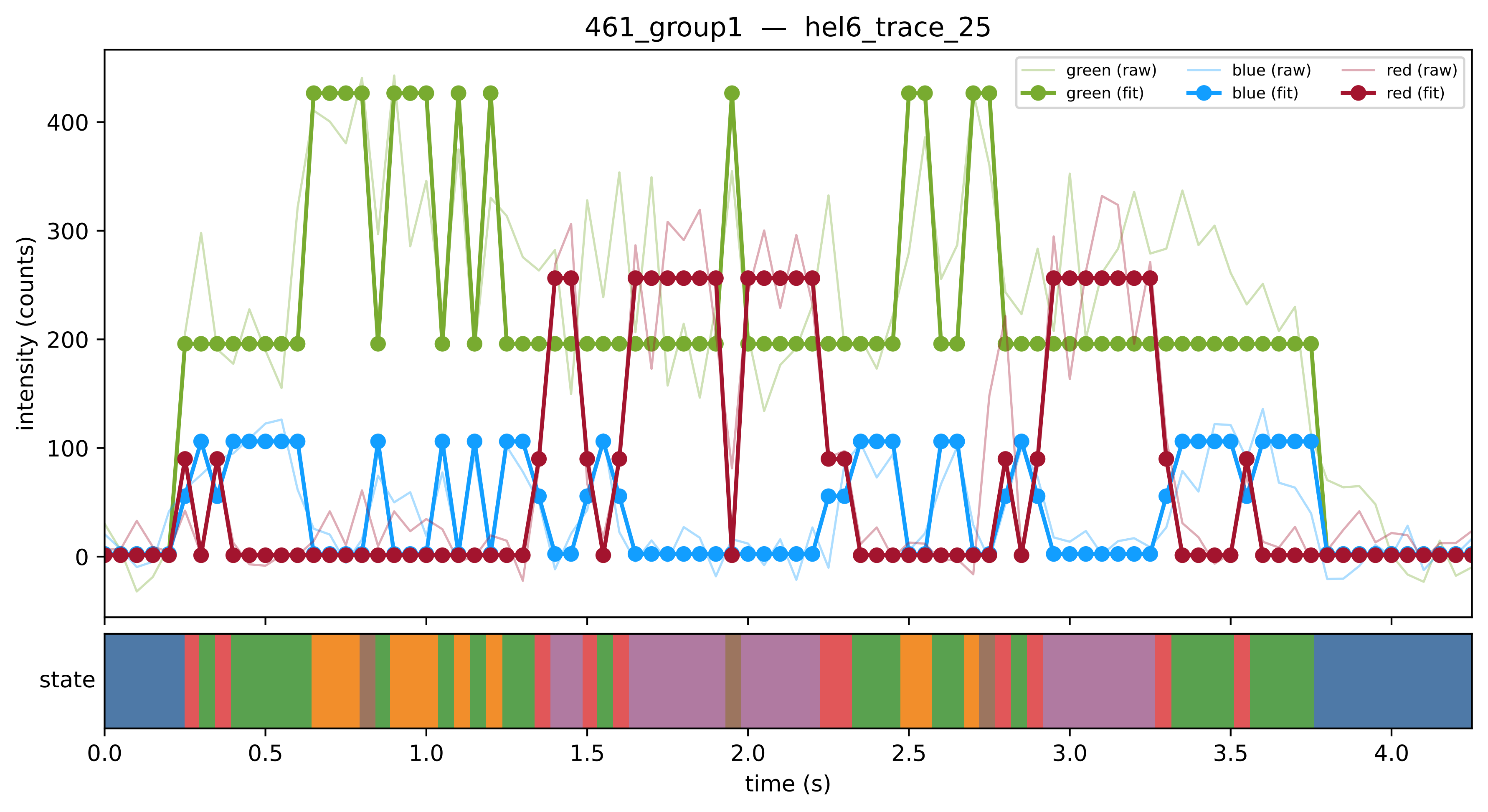
461_group1 — hel6_trace_16
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
100
200
300
400
500
1
2
3
4
5
6
7
8
9

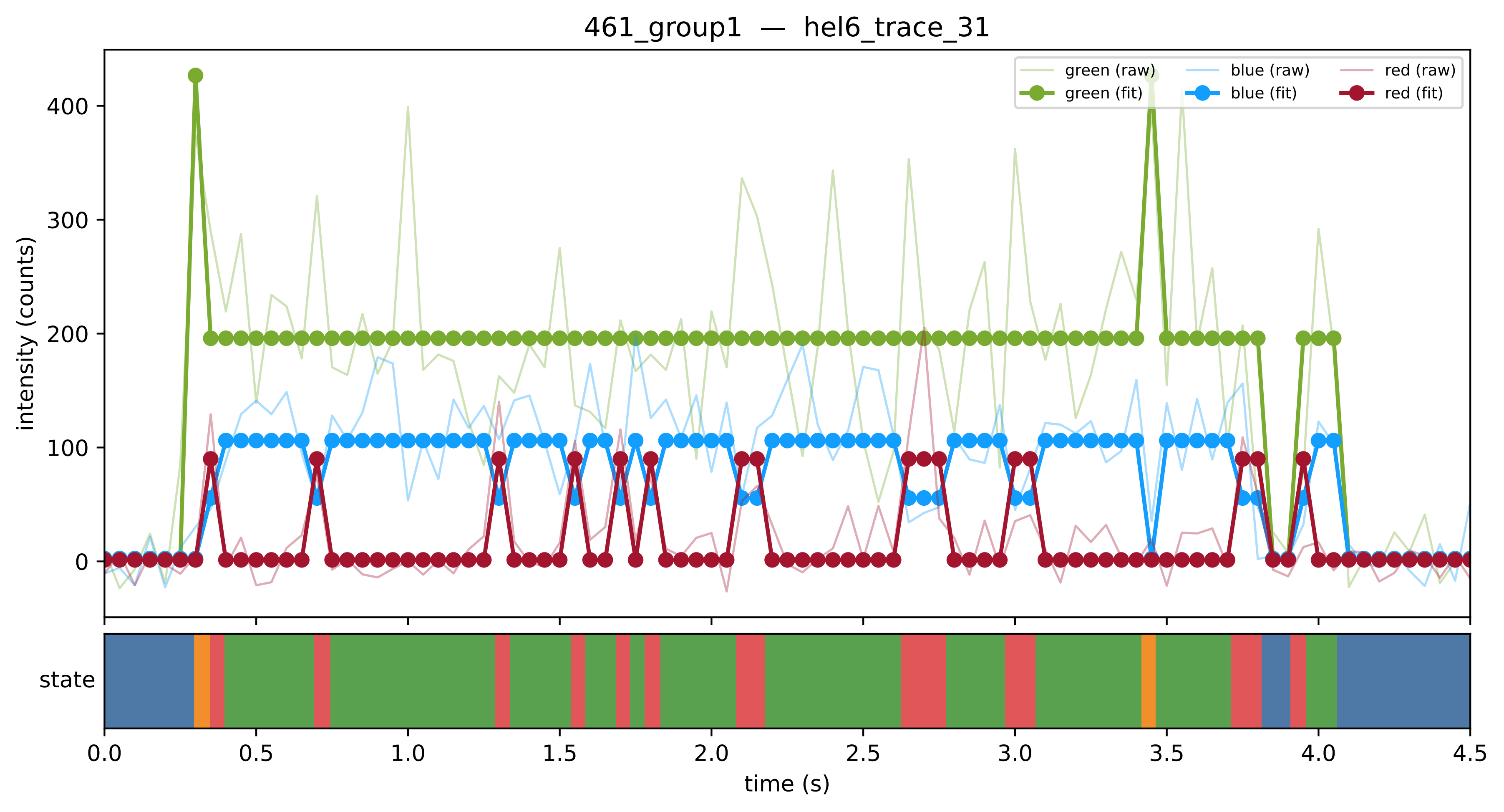


461_group1 — hel6_trace_18
intensity (counts)
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
time (s)
0
100
200
300
400
500
600
700
0.0
0.5
1.0
1.5
2.0
2.5

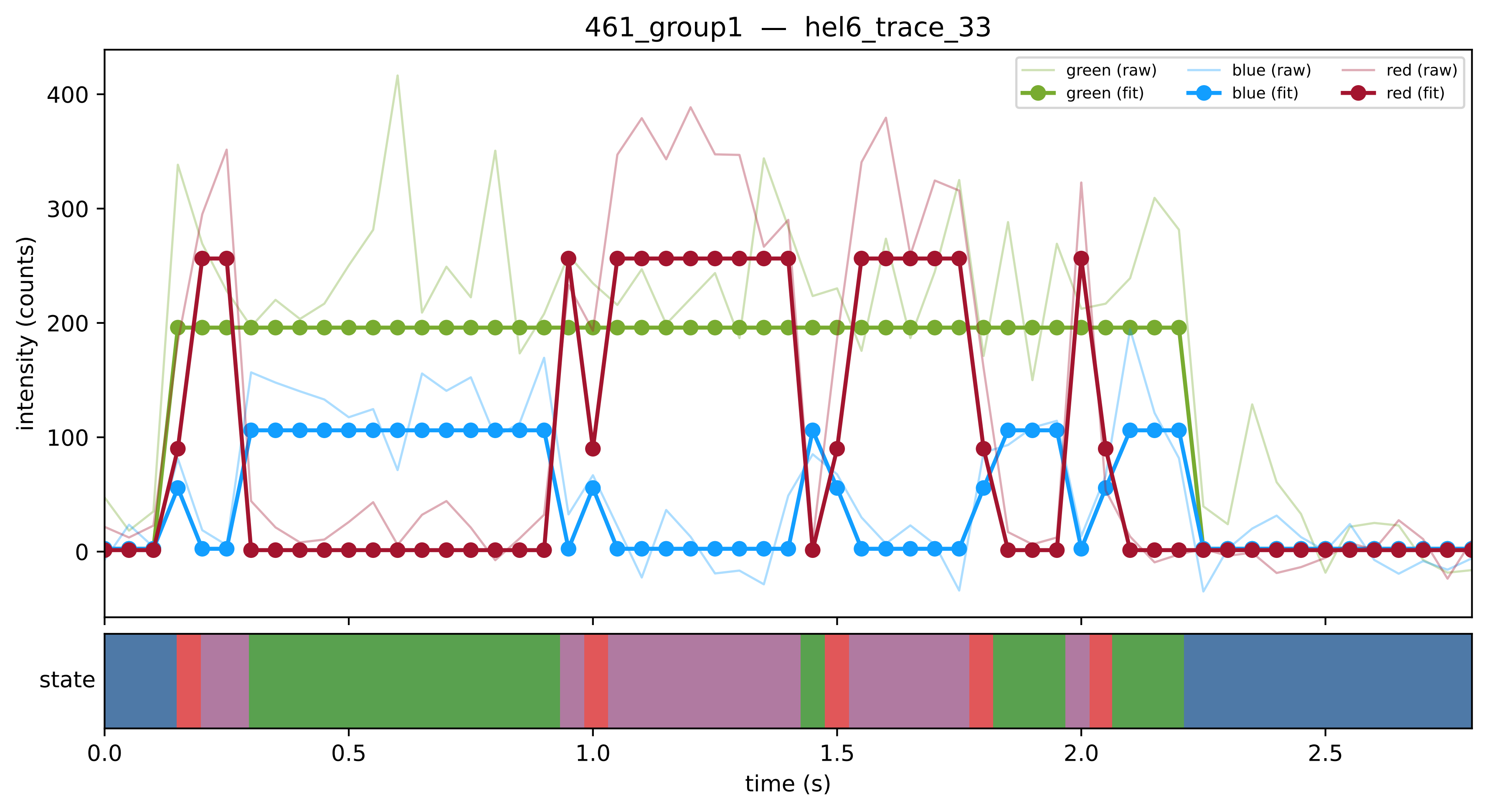


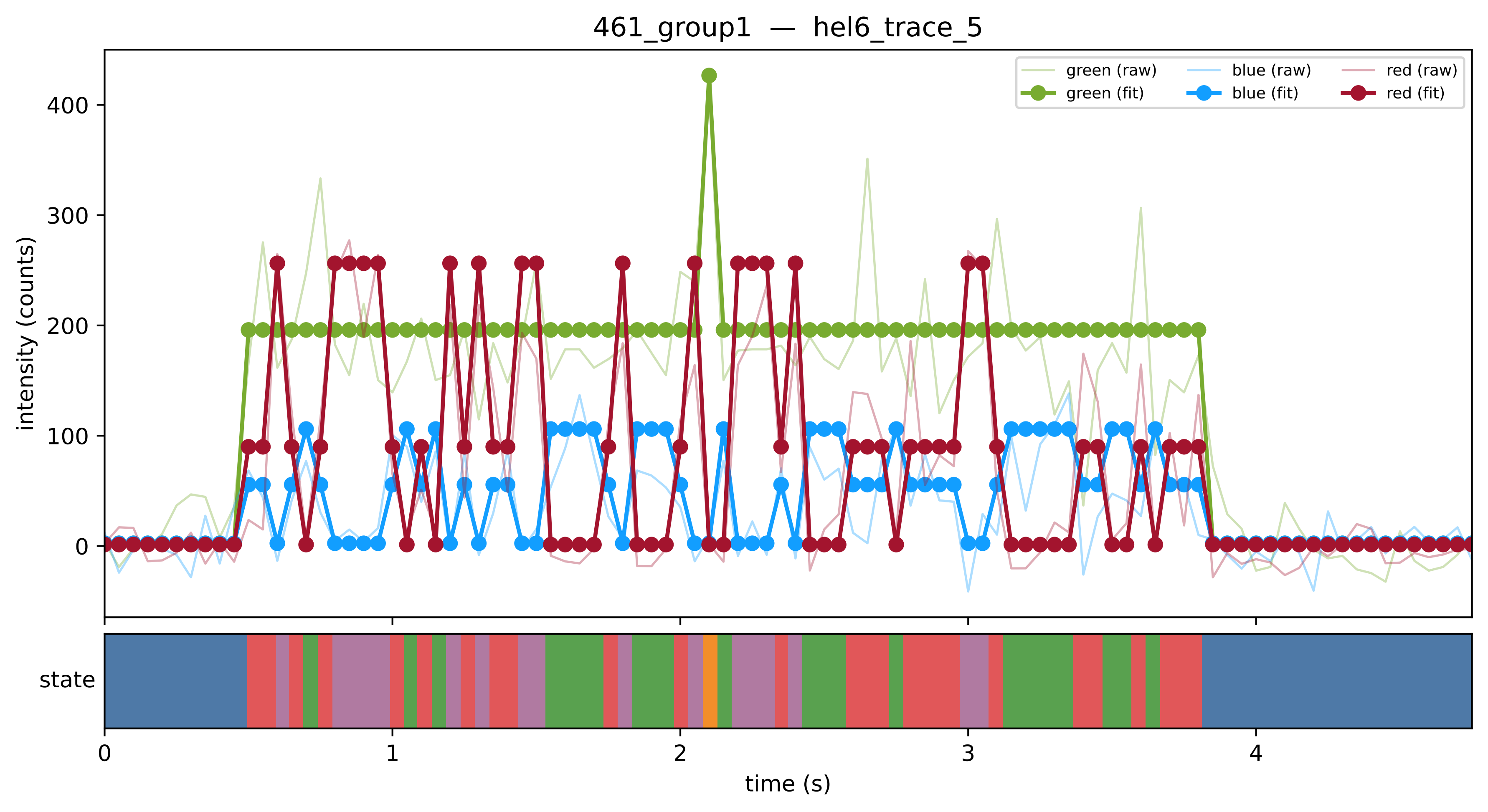
461_group1 — hel6_trace_22
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
500
400
300
200
100
0
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
time (s)

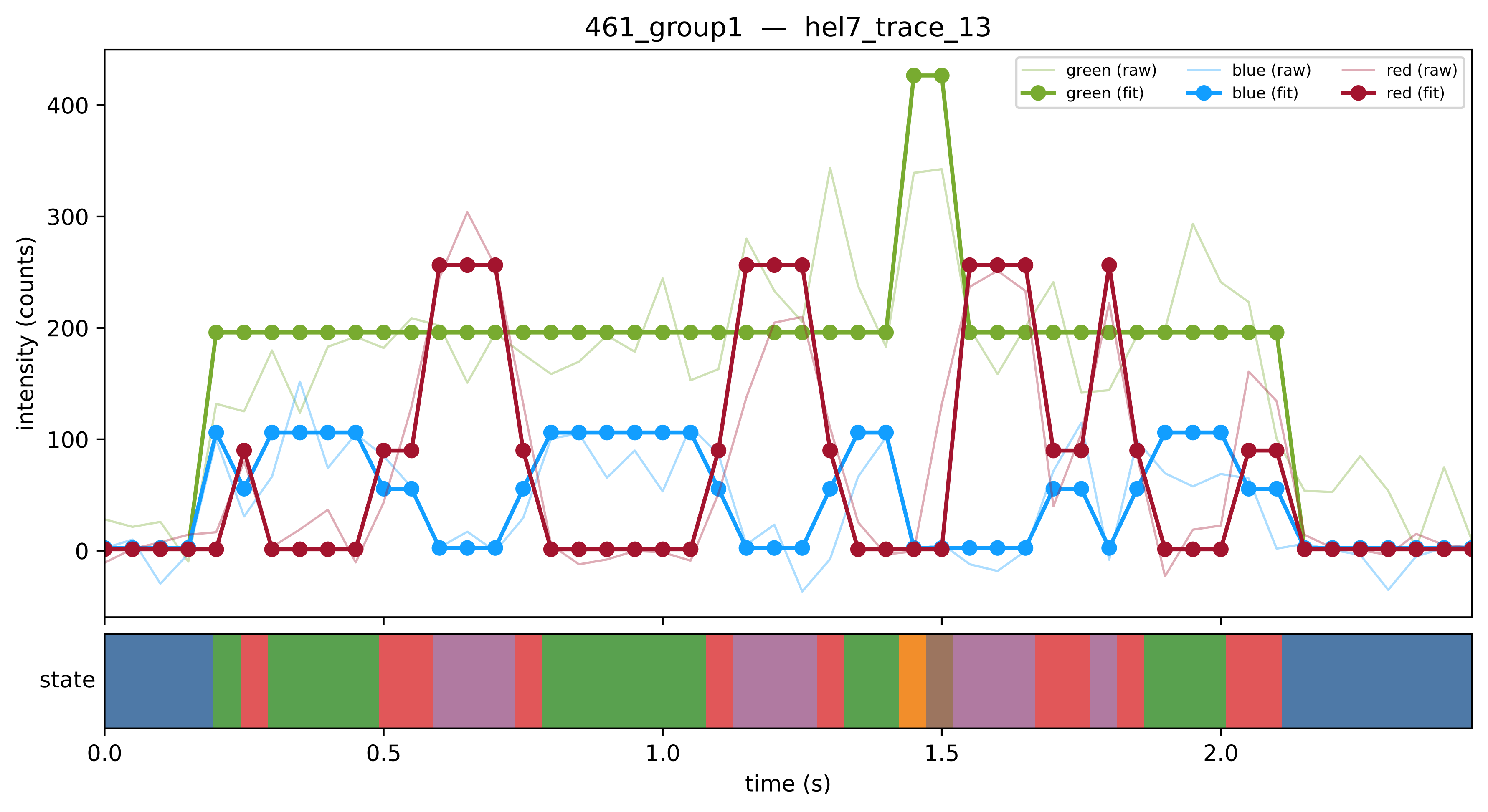

461_group1 — hel6_trace_25
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
time (s)



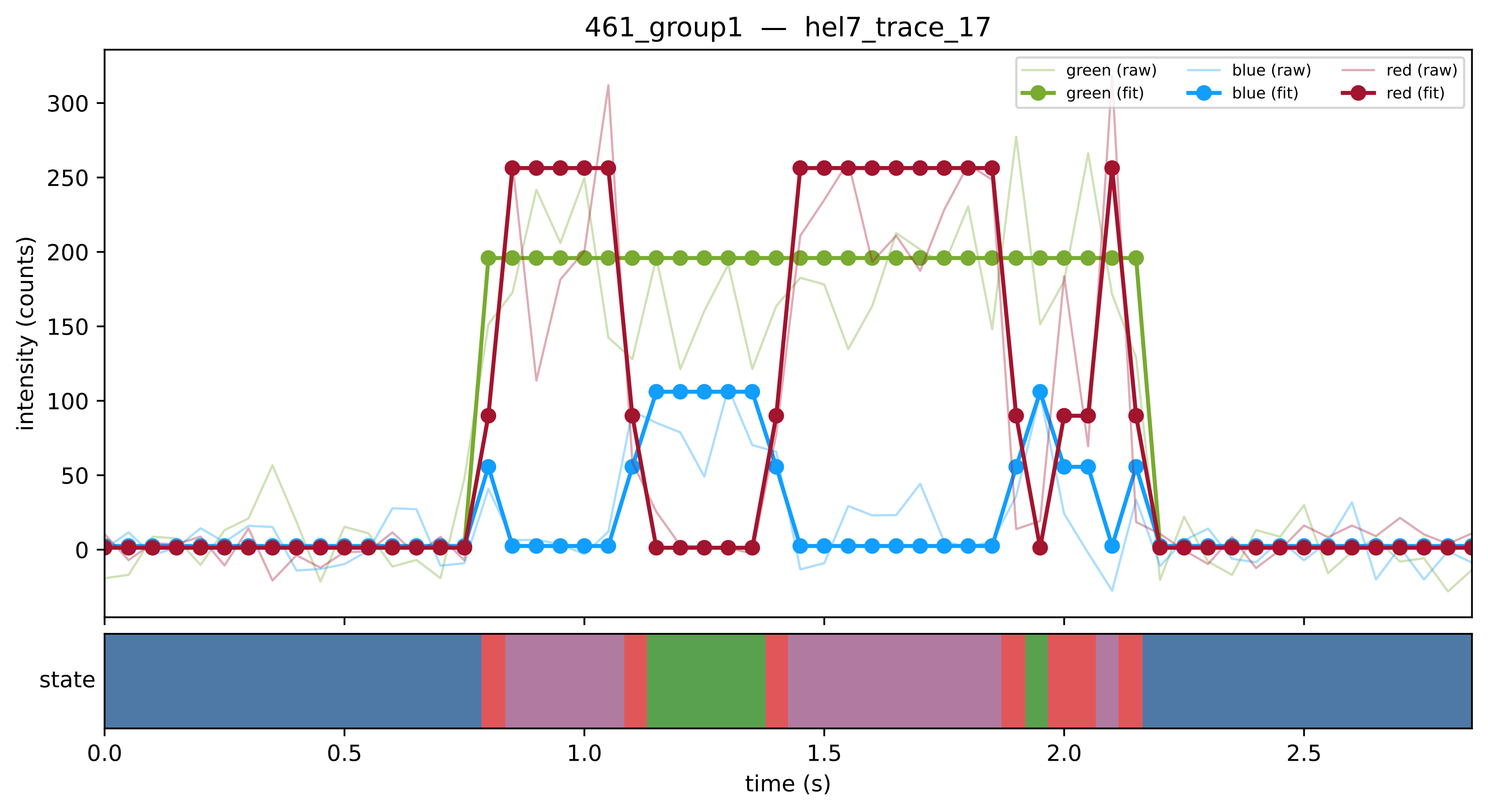
461_group1 — hel6_trace_31
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
time (s)

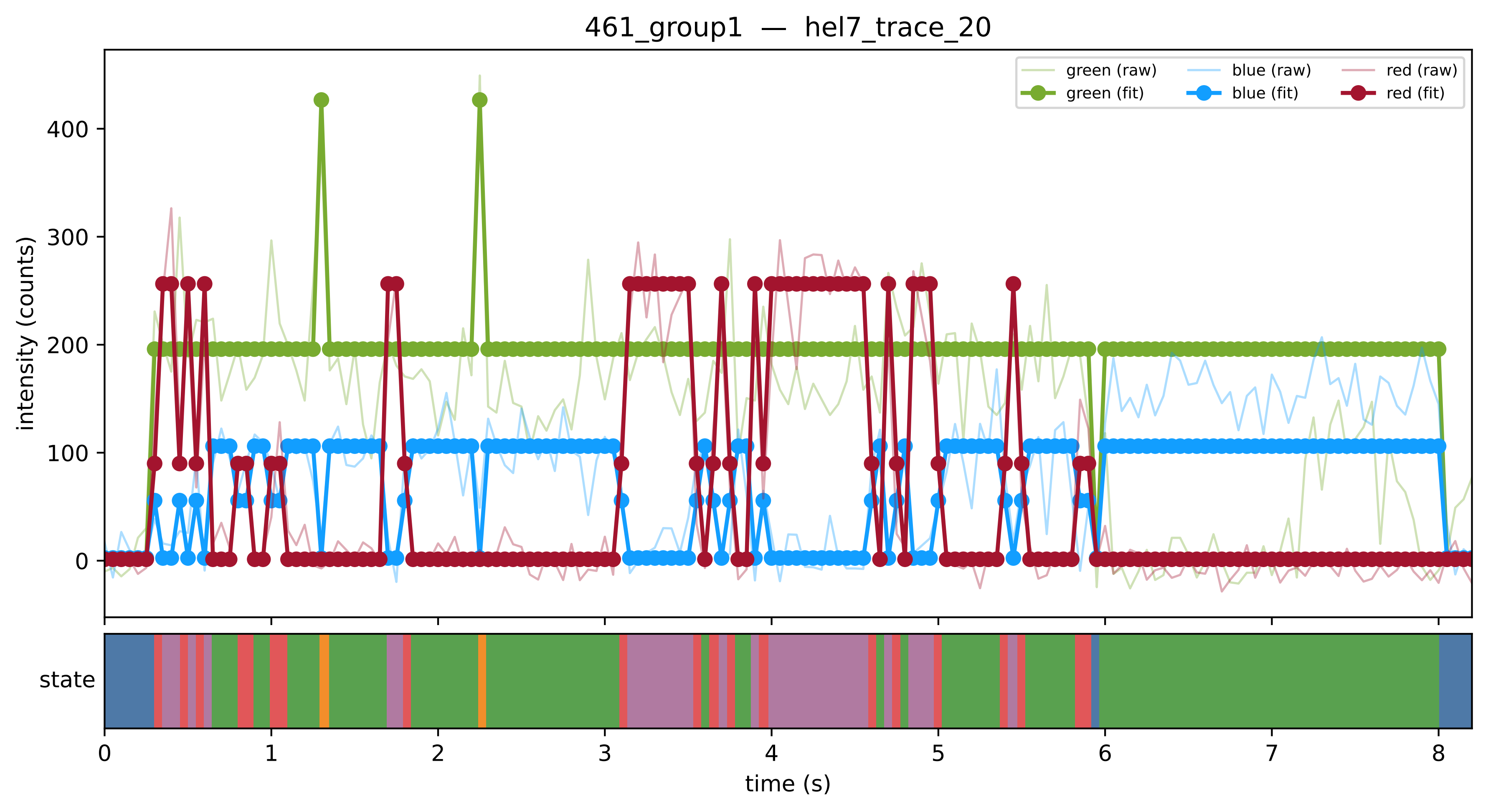

461_group1 — hel6_trace_33
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)


461_group1 — hel6_trace_5
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0
1
2
3
4
time (s)

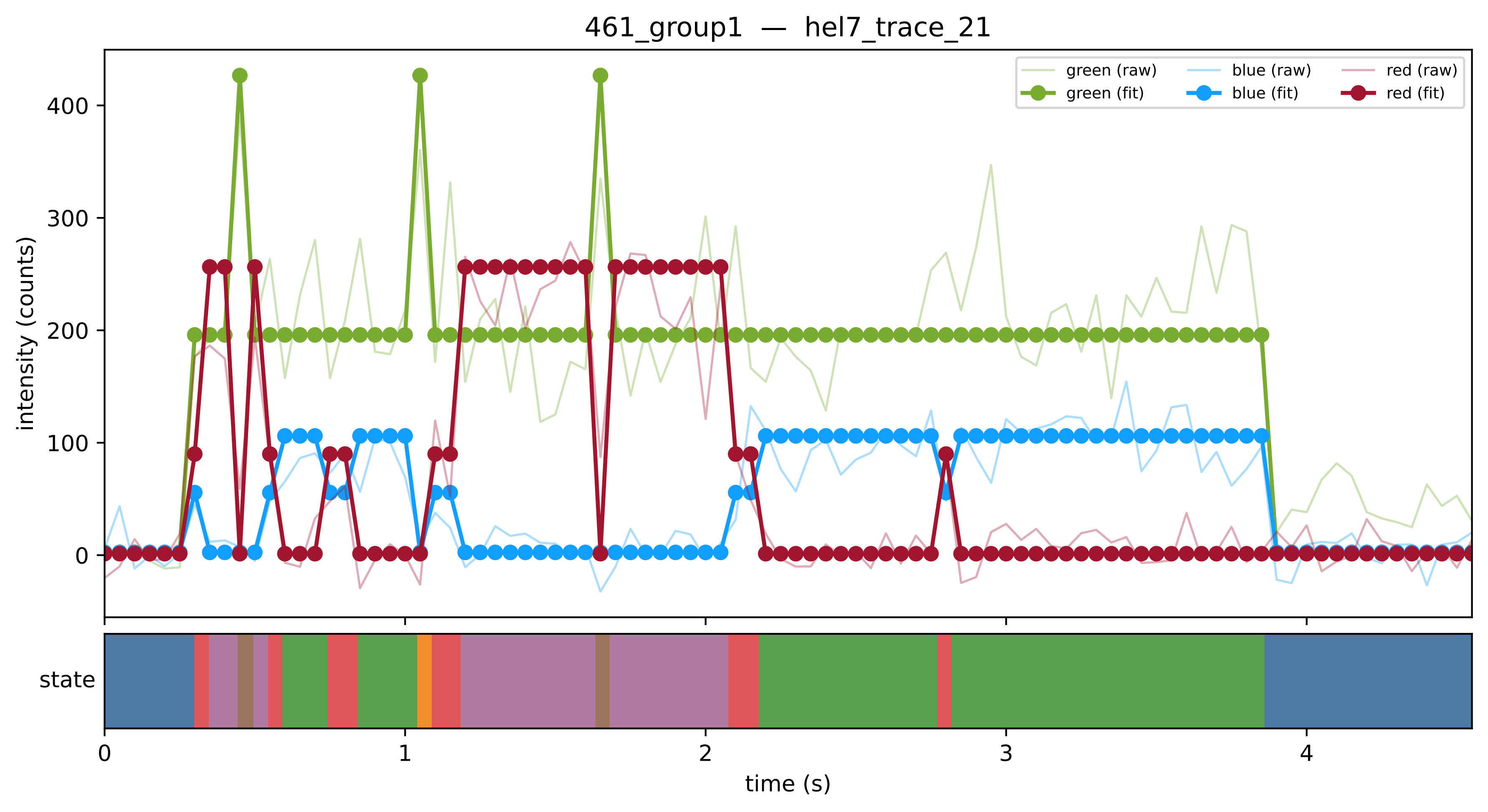


461_group1 — hel7_trace_13
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
400
300
200
100
0
intensity (counts)
state
0.0
0.5
1.0
1.5
2.0
time (s)

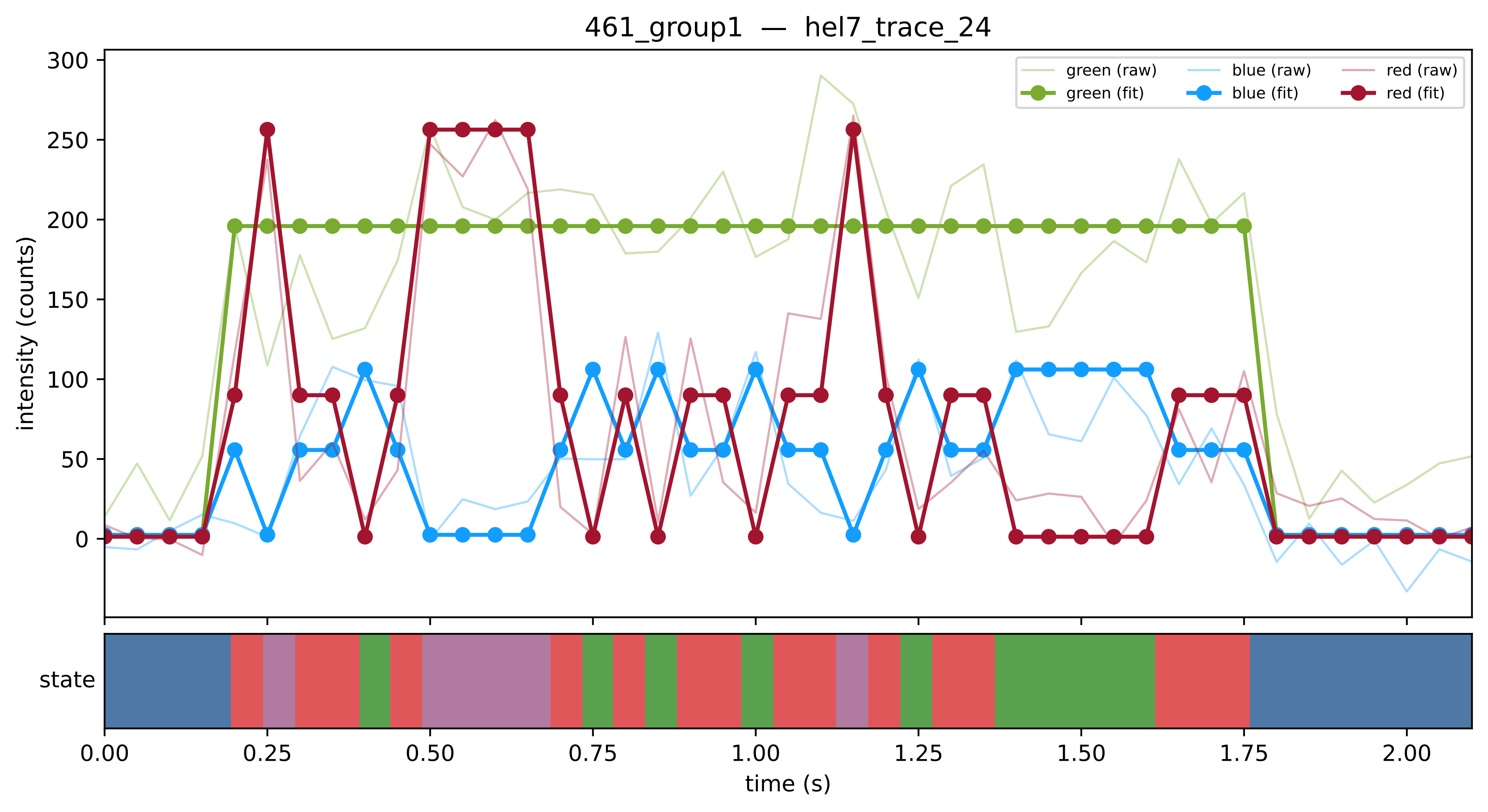

461_group1 — hel7_trace_17
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
50
100
150
200
250
300
state
0.0
0.5
1.0
1.5
2.0
2.5
time (s)



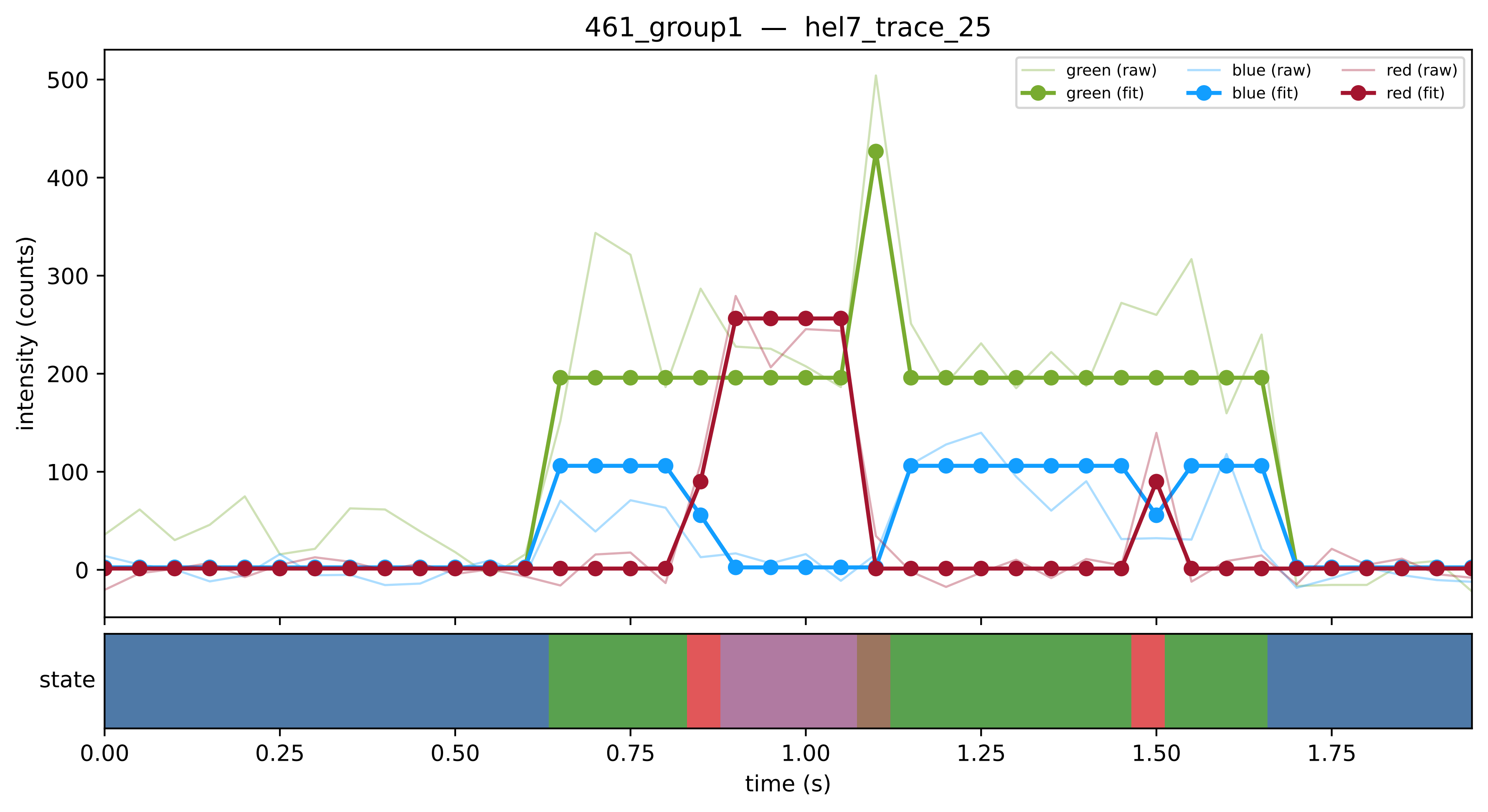
461_group1 — hel7_trace_20
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
state
0
1
2
3
4
5
6
7
8
time (s)

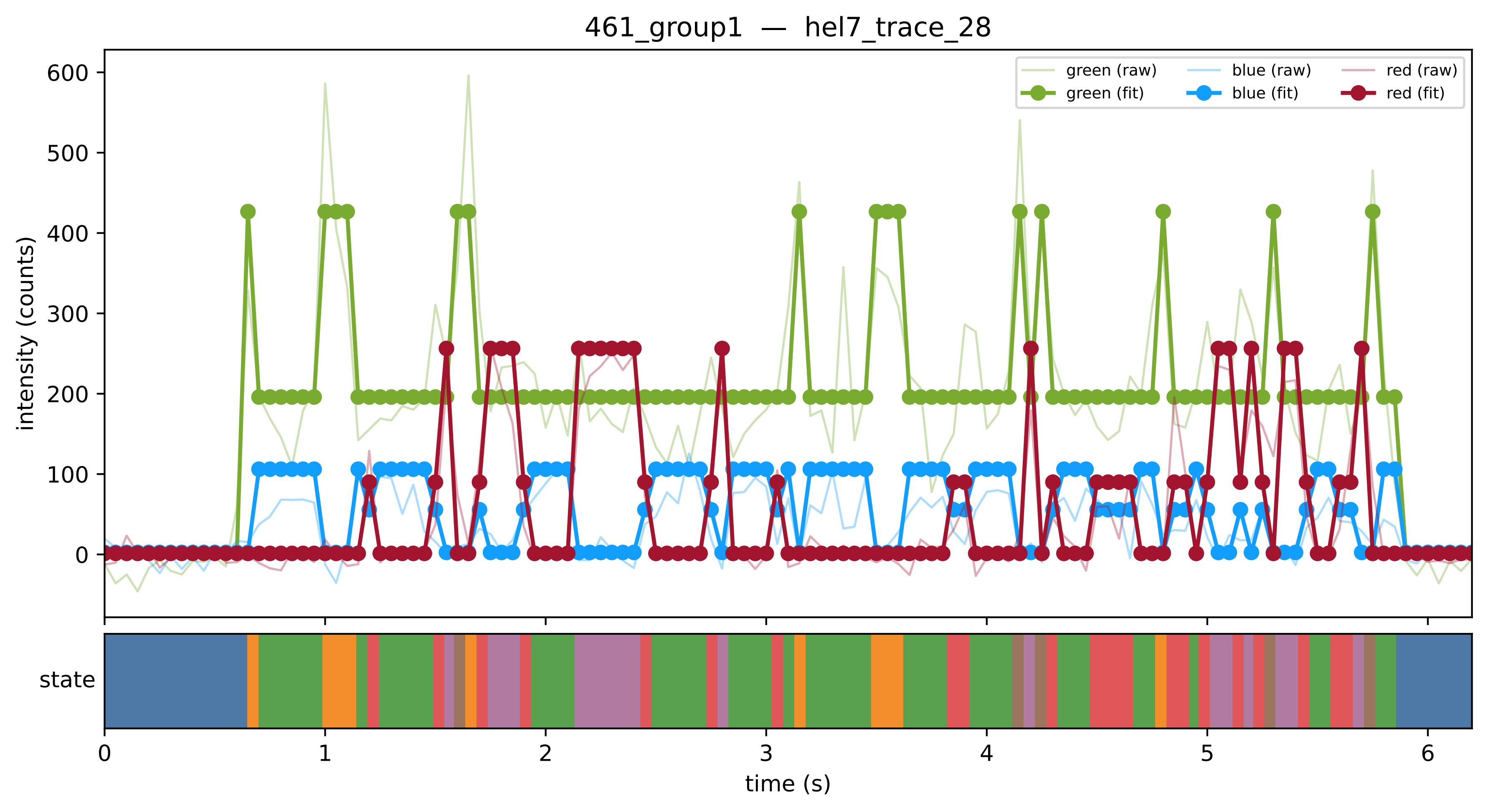


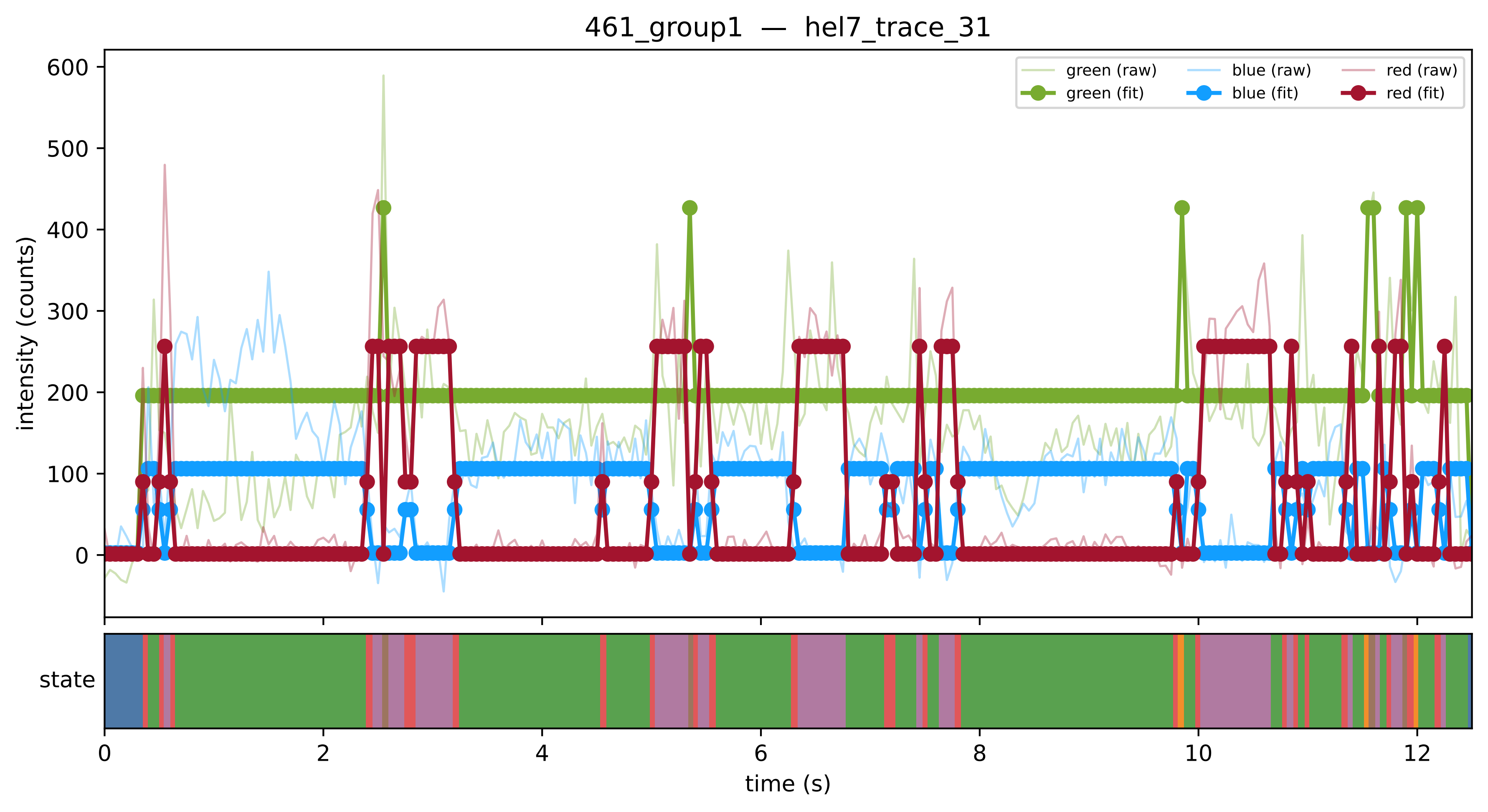
461_group1 — hel7_trace_21
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0
1
2
3
4
time (s)

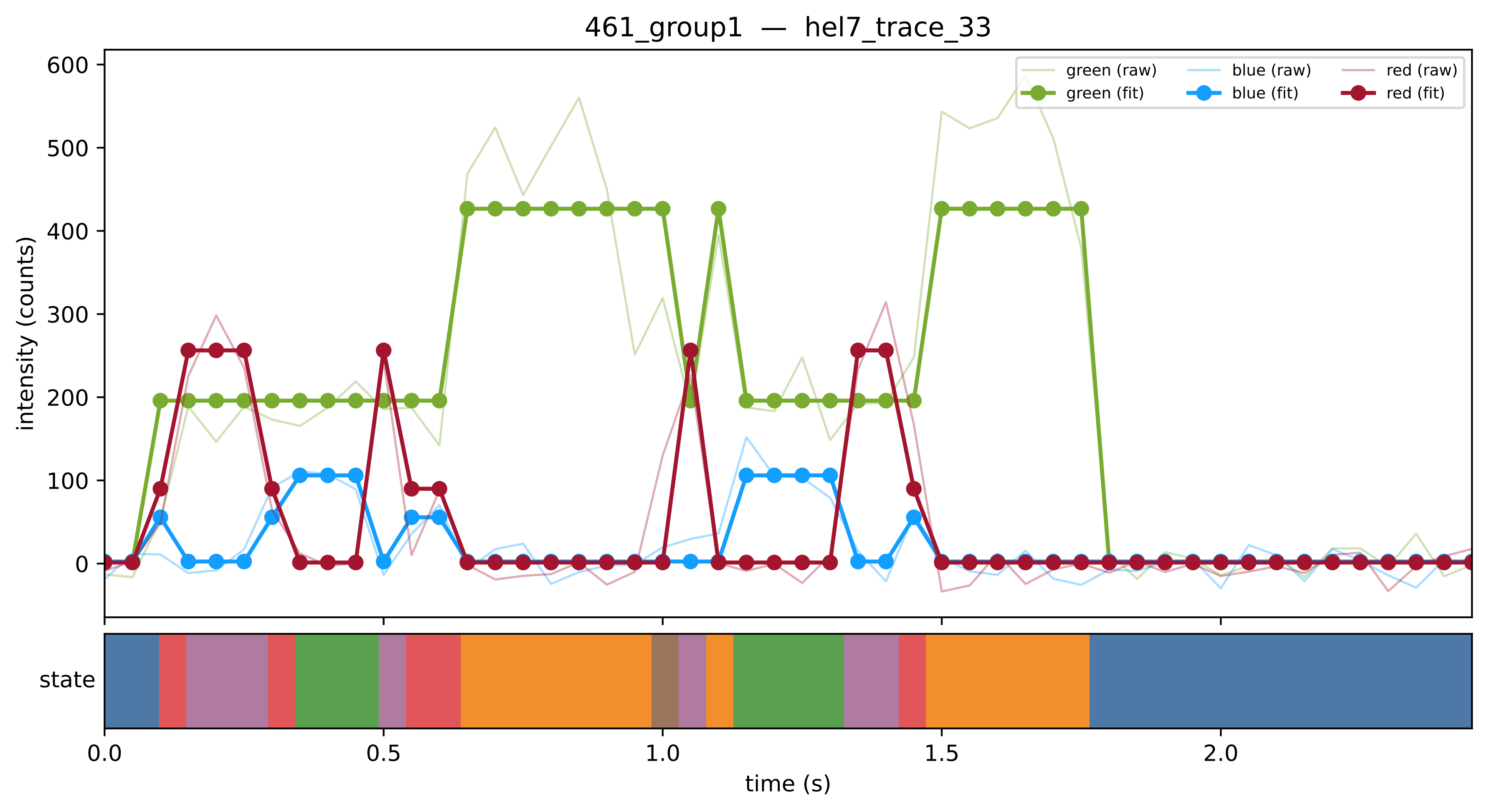


461_group1 — hel7_trace_24
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
intensity (counts)
300
250
200
150
100
50
0
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time (s)

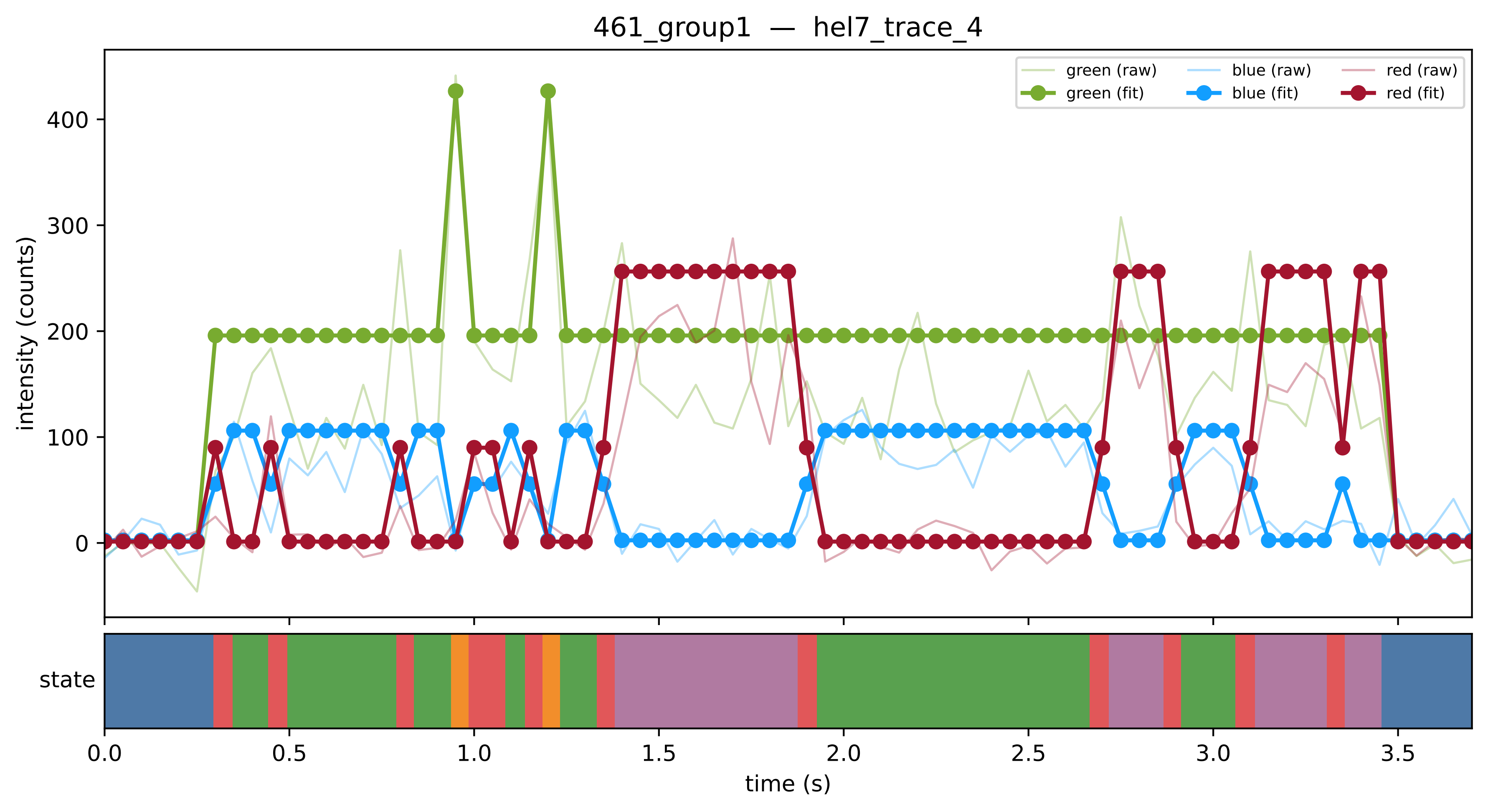

461_group1 — hel7_trace_25
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
0
100
200
300
400
500
state
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
time (s)

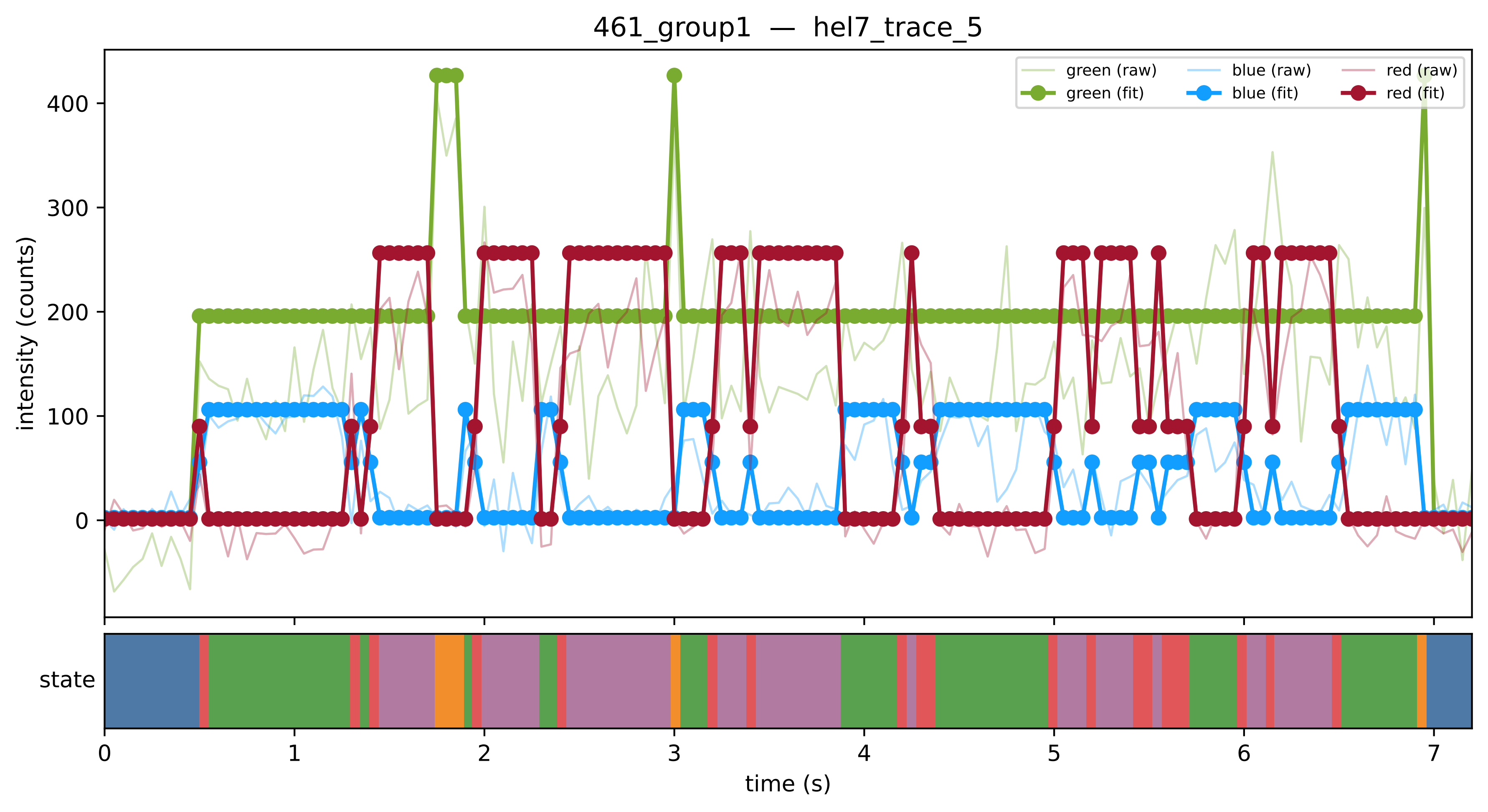

461_group1 — hel7_trace_28
intensity (counts)
600
500
400
300
200
100
0
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0
1
2
3
4
5
6
time (s)


461_group1 — hel7_trace_31
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
600
500
400
300
200
100
0
state
0
2
4
6
8
10
12
time (s)



461_group1 — hel7_trace_33
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
600
500
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
time (s)


461_group1 — hel7_trace_4
green (raw)
blue (raw)
red (raw)
green (fit)
blue (fit)
red (fit)
intensity (counts)
400
300
200
100
0
state
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
time (s)



461_group1 — hel7_trace_5
intensity (counts)
0
100
200
300
400
green (raw)
green (fit)
blue (raw)
blue (fit)
red (raw)
red (fit)
state
0
1
2
3
4
5
6
7
time (s)